Program overview

08-Dec-2022 11:12

Year	**2022/2023**
Organization	**Civil Engineering and Geosciences**
Education	**Bachelor Applied Earth Sciences**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
AES Bachelor 2022			
AES BSc 1e jaar 2022	**AES BSc 1st year 2022**		
AESB1120-15	Principles of Chemistry & Thermodynamics	5	
AESB1120-15 Toets 1	*Exam 1*	*0*	
AESB1120-15 Toets 2	*Exam 2*	*0*	
AESB1120-15 Toets 3	*Practical*	*0*	
AESB1130-21	Geology 1: Basics	5	
AESB1130-21 Exam 01	*General geology the basics*	*3*	
AESB1130-21 Exam 02	*Maps and sections*	*2*	
AESB1211	Mathematics 1	6	
AESB1211 Toets 1	*Analyse deeltentamen 1*	*3*	
AESB1211 Toets 2	*Deeltoets periode 1*	*0*	
AESB1211 Toets 3	*Aanlyse deeltentamen 2*	*3*	
AESB1211 Toets 4	*Deeltoets periode 2*	*0*	
AESB1213	Probability and Statistics	5	
AESB1213 Toets 1	*Lab*	*0*	
AESB1213 Toets 2	*Midterm*	*2,5*	
AESB1213 Toets 3	*Final Exam*	*2,5*	
AESB1230	Geology 2: North West Europe	5	
AESB1230 Toets 1	*North West Europe*	*0*	
AESB1230 Toets 2	*Rock Types*	*0*	
AESB1230 Toets 3	*Structural Geology*	*0*	
AESB1243	Grand Challenges and Applied Earth Sciences	9	
AESB1243 Toets 1	*Grand Challenges Advisory Report*	*4,5*	
AESB1243 Toets 2	*Grand Challenges Presentation*	*1,35*	
AESB1243 Toets 3	*Grand Challenges Exam*	*3,15*	
AESB1320-17	Mechanics	5	
AESB1411	Linear Algebra	5	
AESB1420-17	Electricity and Magnetism	5	
AESB1430-14	Geology 3: geological systems and excursion	5	
AESB1430-14 Toets 1	*Report excursion*	*0*	
AESB1430-14 Toets 2	*Geological systems*	*0*	
AESB1440-21	Methodology of Geophysics and Remote Sensing	5	
AESB1440-21 Exam 01	*Final exam / Lab / Midterm*	*3*	
AESB1440-21 Exam 02	*Final exam / Lab / Midterm*	*1,5*	
AESB1440-21 Exam 03	*Final exam / Lab / Midterm*	*0,25*	
AESB1440-21 Exam 04	*Final exam / Lab / Midterm*	*0,25*	
AES BSc 2e jaar 2022	**AES BSc 2nd year 2022**		
AESB2110-21	Mathematics 4: Differential Equations	5	
AESB2110-21 toets 1	*Final exam*	*5*	
AESB2110-21 toets 2	*Lab: practical*	*0*	
AESB2110-21 toets 3	*Lab: MOOC*	*0*	
AESB2122	Signals and Systems with Python	5	
AESB2122 Toets 1	*Midterm*	*2,5*	
AESB2122 Toets 2	*Final Exam*	*2,5*	
AESB2122 Toets 3	*Lab/practical sessions*	*0*	
AESB2140-21	Geophysical Methods for Subsurface Characterization	5	
AESB2140-21 Toets 1	*Final exam*	*2,5*	
AESB2140-21 Toets 2	*Lab*	*2,5*	
AESB2210-18	Mathematics 5	5	
AESB2210-18 Toets 1	*Practical*	*2*	
AESB2210-18 Toets 2	*Exam*	*3*	
AESB2220-20	Chemical Thermodynamics	5	
AESB2220-20 Toets 1	*Final exam / Lab /Midterm*	*5*	
AESB2220-20 Toets 2	*Assignments*	*0*	
AESB2230	Sedimentology & Reservoir Geology	5	
AESB2230 Toets 1	*Written exam*	*2,5*	
AESB2230 Toets 2	*Practical exam*	*2,5*	
AESB2320	Physical Transport Phenomena	5	
AESB2320 Toets 1	*Exam 1*	*2,5*	
AESB2320 Toets 2	*Exam 2*	*2,5*	
AESB2330	Soil Mechanics	5	
AESB2342	Rock Mechanics	5	

AESB2430	Geological Fieldwork: Data Acquisition	5
AESB2431	Geological Fieldwork: Data Integration	5
AESB2440	Geostatistics & Remote Sensing	5
AESB2440 Toets 1	*Exam*	*2,5*
AESB2440 Toets 2	*Exercises*	*2,5*
AES BSc 3e jaar 2022	**AES BSc 3rd year 2022**	
AES BSc 3e jaar Minor Programma	**AES BSc 3rd year Minor Programma**	
AES BSc 3e jaar Major Programma	**AES BSc 3rd year Major Program**	
AESB3340	Mechanics and Transport by Flow in Porous Media	5
AESB3341-18	Petrophysics & Image Analysis	5
AESB3341-18 Toets 1	*Exam*	*1*
AESB3341-18 Toets 2	*Lab report*	*2*
AESB3341-18 Toets 3	*Image Analysis*	*2*
AESB3343	Mineral Resource Geology and Modelling	5
AESB3343 Toets 1	*Assignments*	*1*
AESB3343 Toets 2	*Exam*	*2*
AESB3343 Toets 3	*Project Report*	*2*
AESB3400	Bachelor Thesis	10
AESB3440	Field Exploration Project	5

Year	**2022/2023**
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AES Bachelor 2022	
Introduction 1	For more information, please check the regulations: http://studenten.tudelft.nl/en/students/faculty-specific/ceg/regulations-ter-rules-and-guidelines/

Year	**2022/2023**
Organization	**Civil Engineering and Geosciences**
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AES BSc 1e jaar 2022

AESB1120-15	Principles of Chemistry & Thermodynamics	5
Responsible Instructor	Prof.dr. J.H. van Esch	
Course Coordinator	Prof.dr.ir. T.J. Heimovaara	
Instructor	Dr. R. Eelkema	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course is divided into a theoretical and a practical part, and scheduled parallel during the quarter. In the theoretical part of the course, based upon a brief repetition of high school chemistry, we deepen several concepts in chemistry and thermodynamics. See also the outline as described below. The practical part consists of the modelling of chemical equilibria and experimental data by computational tools. The computer practicals are aimed to connect theoretical knowledge with experiment. In the theoretical part the following subjects are covered: Matter, Measurement, and Problem Solving Atoms and Elements Molecules, Compounds, and Chemical Equations Chemical Quantities and Aqueous Reactions Gases, Thermochemistry The Quantum-Mechanical Model of the Atom Periodic Properties of the Elements Chemical Bonding: Molecular Shapes, Valence Bond and Molecular Orbital Theory Liquids, Solids, and Intermolecular Forces Solutions Chemical Kinetics Chemical Equilibrium Free energy and thermodynamics Acids and Bases Aqueous Ionic Equilibrium Electrochemistry	
Study Goals	The objective of this course is to make the student familiar with the main principles of chemistry and thermodynamics and to apply the main principles of chemistry in solving basic chemical problems. [Knowledge] At the end of this course, the student is able to: - give a qualitative quantum mechanical description of the atom - deduce the basic properties of elements from the periodic table - give a qualitative description of the basic properties of solids, liquids and gases - give a qualitative description of the structure and properties of metals, salts, molecules, and ionic and molecular solutions - recognize and describe basic chemical reactions (redox, acid-base, precipitation) - can qualitatively relate intermolecular forces to properties of solids and liquids - formulate chemical equilibria and predict how they respond to change - calculate composition, equilibrium constant, and free energy using the relations between chemical equilibria, free energy and enthalpy and entropy - carry out simple calculations of composition, equilibrium constant, and electrochemical potential using the Nernst relation. - carry out simple structure-property relations for organic molecules. - be able to calculate the ionic strength and apply activity corrections with the Davies equation. [Skills] At the end of this course, the student is able to: - Classify matter - Apply the law of conservation of mass - Work with chemical units of quantity and mass in solids, liquids and gases - Predict qualitatively and quantitatively the product of basic chemical reactions - Apply simple enthalpy calculations to deduce if reactions are endo- or exothermic - Predict properties of elements from the periodic table - Use Valence Bond theory to describe molecular structure - Calculate chemical equilibria from the concentrations and vice versa for acid-base, redox, and precipitation reactions in solution and gas-phase reactions - Apply simple structure-property relationships for elements, ions, and molecules to predict or explain their chemical and physical behavior - Implement and interpret the results of simple aqueous reaction problems including precipitation and exchange with the gas phase in PHREEQC [Attitude] At the end of this course, the student is familiar with: - application of a model to describe chemical experiments - application of a scientific method for problem solving - relation between theory, model and experiment - uncertainties and limitations of experiments - reporting on simple modelling of data from chemical expriments - drawing conclusions based on (experimental) evidence	
Education Method	The course will consist of lectures and exercises, using a text book and an electronic learning environment, and modelling aqueous reactions using the software PHREEQC. Each lecture will cover one particular topic and is accompanied by an exercise hour. From students it is expected that they have studied the lecture topic from the book in advance of the lecture, and complete the assigned exercises for each lecture before start of the next lecture. For the PHREEQC practicals it is expected that the students have studied the applicable theory, before each practical. The students perform three experiments. Each PHREEQC practical is accompanied by exercise hours regarding the required theory. For the third PHREEQC practical a set of questions and a report have to be made. Students are expected to hand in the report within one week after finishing the last PHREEQC practical.	
Course Relations	Chemistry and Thermodynamics 2	
Assessment	- Theoretical part of the course: 2 written interim summative tests. - Practical part of the course: 1 written report including answers to provided questions of the final assignment.	

	Re-exam: - practical part of the course: redo a different final assignment and rewrite the report. The maximum grade for the re-exam of the practical is a 6. - theoretical part of the course: if one of the intermediate tests is below 5,0, a re-exam for this test is needed. There is a resit in week 9 and/or 10 of period 4 in which the division of the intermediate tests is denoted. Each combination of intermediate tests can be done. The final grade is built up as follows: - a grade will be given for each of the written tests; - one final grade will be given for the practical part on PHREEQC. The final mark results from the average of the two tests and the practical mark (equivalent in weight).
Remarks	5 EC or 140 hrs, divided as follows: 80 hours for lectures and practicals, 56 hours for self study, 4 hours assessment(theory): Lectures 28hrs (6 weeks) Exercise hours 28 hrs (6 weeks) Practicum 12hrs (3weeksx1dayx4hrs) Reporting 12 hrs (3weeksx1dayx4hrs) Assessment 4 hrs (2 tests of 2hr) Self-study 56 hrs (theory and reports; 9 weeks, 6hrs)
Contact	Prof. dr. J.H. van Esch Advanced Soft Matter, Department of Chemical Engineering, Delft University of Technology van der Maasweg 9, 2629 HZ Delft, The Netherlands Phone +31 15 27 88826 Fax +31 15 27 84289 Email j.h.vanesch@tudelft.nl
Expected prior Knowledge	Chemistry at VWO level
Academic Skills	Reading, Problem solving, Analysis, Writing
Literature & Study Materials	Principles of Chemistry: A molecular Approach: international Edition, 2/E, Nivaldo J. Tro
Judgement	The final grade is built up as follows: - a grade will be given for each of the written tests; - one final grade will be given for the practical part on PHREEQC. The final mark results from the average of the two tests and the practical mark (equivalent in weight). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Pen or pencil, paper, calculator. The necessary tables will be provided at the exam. The use of own books with tables (e.g. BINAS) is not allowed.
Collegerama	No

AESB1120-15 Toets 1	***Exam 1***	***0***
Responsible Instructor	Prof.dr. J.H. van Esch	
Contact Hours / Week x/x/x/x	see AESB1120-15	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	see AESB1120-15	
Study Goals	see AESB1120-15	
Education Method	see AESB1120-15	
Assessment	see AESB1120-15	
Expected prior Knowledge	see AESB1120-15	
Academic Skills	see AESB1120-15	
Literature & Study Materials	see AESB1120-15	
Judgement	see AESB1120-15	
Permitted Materials during Exam	see AESB1120-15	
Collegerama	No	

AESB1120-15 Toets 2	Exam 2	0
Responsible Instructor	Prof.dr. J.H. van Esch	
Contact Hours / Week x/x/x/x	see AESB1120-15	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	see AESB1120-15	
Study Goals	see AESB1120-15	
Education Method	see AESB1120-15	
Assessment	see AESB1120-15	
Expected prior Knowledge	see AESB1120-15	
Academic Skills	see AESB1120-15	
Literature & Study Materials	see AESB1120-15	
Judgement	see AESB1120-15	
Permitted Materials during Exam	see AESB1120-15	
Collegerama	No	

AESB1120-15 Toets 3	Practical	0
Responsible Instructor	Prof.dr. J.H. van Esch	
Contact Hours / Week x/x/x/x	see AESB1120-15	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	see AESB1120-15	
Study Goals	see AESB1120-15	
Education Method	see AESB1120-15	
Assessment	see AESB1120-15	
Expected prior Knowledge	see AESB1120-15	
Academic Skills	see AESB1120-15	
Literature & Study Materials	see AESB1120-15	
Judgement	see AESB1120-15	
Permitted Materials during Exam	see AESB1120-15	
Collegerama	No	

AESB1130-21	Geology 1: Basics	5
Responsible Instructor	Dr. G. Bertotti	
Instructor	Drs. J.C. Blom	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Parts	The Geology 1 course is composed of two components dedicated to 1) general knowledge of the system Earth, 2) tools for the 3D geometric representation of geological objects.	
Course Contents	General knowledge of the system Earth, tools for the 3D geometric representation of geological objects. Part 1: the rock cycle, the structure and functioning of the Earth and its outer layer, the lithosphere; horizontal deformations and vertical movements; the climate system; notions of hydrogeology; sedimentary systems from the source to the sink (marine systems).(Bertotti) Part 2: focuses on tools to represent the architecture of geologic bodies on maps and how these can be used to infer the geological evolution of a specific region (Blom)	
Study Goals	The goal of the Geology I course is to provide students basic knowledge and skills to understand relevant geological processes and enable them to make geological predictions required by their future careers. Knowledge: Students will be able to: - Describe the first order structure and functioning of the Earth. This includes basic knowledge of i) elements, minerals and rocks, ii) of the internal structure of the Earth with particular attention to the lithosphere and the crust, and iii) of plate tectonics. - Describe the main characteristics of sedimentary, igneous and metamorphic rocks and will have an understanding of how these rocks were formed. - Describe kinematics and basic physics of vertical movements (subsidence creating sedimentary basins and uplift creating relief) and horizontal deformations at lithospheric to thin section scale. They will be able to quantify simple relations between relief and crustal structure. They will be able to extract deformation scenarios from simple geological documents such as geological and seismic sections. - Describe exogene processes and their impact on the sedimentary record. Processes include climate, terrestrial water flow and oceanography. - Describe the main sedimentary environments and their evolution through time. Students will also be able to infer such evolutionary schemes from simple geological records such as stratigraphic columns and use them for simple paleogeographic reconstructions. Skills. Students will be able to: - Interpret geologic maps, geologic sections and other basic documents such as seismic section to infer the 3D architecture of the area/volume of investigation - Use the techniques mentioned above to constrain larger scale processes and make predictions for spatially and temporally unknown parts of the Earth system (mainly Earth upper crust).	
Education Method	The course Geology 1 is a combination of lectures and laboratory work with practicals with maps, physical samples etc. In total there will be 28hrs of lectures, 35hrs of practicals, 4,5hrs of tests and 3hrs for the repetition.	
Assessment	The final grade results from two components: 1) General geology, the Basics (60%) and 2) Geological sections and maps (40%). The grade of component 1 is composed of the results of the practical and of the exam. The percentages will be defined at the beginning of the course The grade of component 2 (Geological maps and sections) is given on the basis of the exam. A grade of 5 or more required for each of the two components. Independent resits for the two components will be held	
Permitted Materials during Tests	To be specified in the lectures before tests.	
Contact	Prof. Dr. G. Bertotti G.bertotti@tudelft.nl tel: +31 (0)15 27 86033	
Expected prior Knowledge	none - introductory course	
Academic Skills	understanding and reading English texts; use basic mathematics tools and physical concepts	
Literature & Study Materials	The book adopted for the Geology 1 course is Understanding Earth (Grotzinger and Jordan) Handouts of lectures will be available on Brightspace	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Coloured pencils, rulers and scales	
Collegerama	No	

AESB1130-21 Exam 01	*General geology the basics*	*3*
Responsible Instructor	Dr. G. Bertotti	
Instructor	Drs. J.C. Blom	
Course Contents	This component is composed of two parts: i) the exam which will focus on the General part of the course and, ii) the results of the practical	
Study Goals	To test the knowledge and skills acquired by the students during the course	
Education Method	see course description	
Assessment	The grade assigned to "exam 01" is made up of two components: the results of the practical and of the exam. The percentage of the two components will be specified at the beginning of the course A minimum grade of 5 is required for both components	
Expected prior Knowledge	see course	
Academic Skills	see course	
Literature & Study Materials	see course	
Judgement	to be specified during the course	
Permitted Materials during Exam	to be specified during the course	

AESB1130-21 Exam 02	*Maps and sections*	*2*
Responsible Instructor	Dr. G. Bertotti	
Instructor	Drs. J.C. Blom	
Course Contents	Refer to AESB1130	
Study Goals	Refer to AESB1130-21	
Education Method	see course	
Assessment	The grade will be assigned on the basis of the exam. The exam is built around a number of exercises	
Expected prior Knowledge	see course	
Academic Skills	see course	
Literature & Study Materials	see course	
Judgement	to be specified during course	
Permitted Materials during Exam	Pens, pencils, pocket calculator	

AESB1211	Mathematics 1	6
Responsible Instructor	Dr. I.M. Smit	
Course Coordinator	Ir. D. den Ouden-van der Horst	
Course Coordinator	Dr. E. Emsiz	
Course Coordinator	Dr. I.M. Smit	
Instructor	Dr. D. Balagué Guardia	
Instructor	Dr. N.D. Verhulst	
Instructor	Dr. E. Emsiz	
Instructor	Y. Murakami	
Instructor	Dr. M.E.L. Jones	
Instructor	Dr. I.M. Smit	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	Part 1: First and second order differential equations. Part 2: Functions of multiple variables.	
Study Goals	After this course you will be able to 1. use differential equations to solve concrete problems, 2. interpret a direction field, 3. solve first order differential equations, 4. use differential equations to model vibrating springs, 5. solve second order linear differential equations, 6. solve differential equations using Taylor polynomials, 7. work with parametric and Cartesian equations of geometric figures, 8. compute the dot product of two vectors, 9. determine the domain of a function of multiple variables and compute its partial derivatives, 10. determine the gradient of a function and solve optimization problems, 11. compute multiple integrals, 12. use cilindrical and spherical coordinates.	
Education Method	Classes on campus with a mixture of lecture and exercise time. Planning may change depending on government and TU regulations.	
Assessment	There will be a midterm in each quarter and two exams, one after the first quarter and one after the second quarter. The midterms and exams will be on campus, either on paper or digital, or possibly a combination of these. Judgement: If your score on the homework assignments is sufficient during a quarter, your result for that part of the course will be the maximum of your grade on the exam and 0.15Midterm+0.85Exam. If this is not the case, the result for that part is the grade that you obtained on the exam. The aforementioned bonus rule is not applicable to resit exams. Your final result is the average of the grades you obtained for part one and part two. To pass this course a student has to score at least 5 points for both part one and part two. (Note: The midterms will not be taken into account during the resits.)	
Expected prior Knowledge	VWO Mathematics B	
Academic Skills	1. Mathematical modelling. 2. Interpret a model. 3. Be critical when performing computations. 4. Appreciate mathematics.	
Literature & Study Materials	Calculus, Volume 1, 2 en 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open book, which can be accessed (for free) through Brightspace.	
Judgement	If your score on the homework assignments is sufficient during a quarter, your result for that part of the course will be the maximum of your grade on the exam and 0.15Midterm+0.85Exam. If this is not the case, the result for that part is the grade that you obtained on the exam. The aforementioned bonus rule is not applicable to resit exams. Your final result is the average of the grades you obtained for part one and part two. To pass this course a student has to score at least 5 points for both part one and part two. (Note: The midterms will not be taken into account during the resits.) For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Analysis formula sheet.	
Collegerama	No	

AESB1211 Toets 1	Analyse deeltentamen 1	3
Responsible Instructor	Dr. I.M. Smit	
Contact Hours / Week x/x/x/x	See AESB1211	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See AESB1211	
Study Goals	See AESB1211	
Education Method	See AESB1211	
Assessment	See AESB1211	
Expected prior Knowledge	See AESB1211	
Academic Skills	See AESB1211	
Literature & Study Materials	See AESB1211	
Judgement	See AESB1211	
Permitted Materials during Exam	See AESB1211	
Collegerama	No	

AESB1211 Toets 2	Deeltoets periode 1	0
Responsible Instructor	Dr. I.M. Smit	
Contact Hours / Week x/x/x/x	See AESB1211	
Education Period	1 2	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	See AESB1211	
Study Goals	See AESB1211	
Education Method	See AESB1211	
Assessment	See AESB1211	
Expected prior Knowledge	See AESB1211	
Academic Skills	See AESB1211	
Literature & Study Materials	See AESB1211	
Judgement	See AESB1211	
Permitted Materials during Exam	See AESB1211	
Collegerama	No	

AESB1211 Toets 3	Aanlyse deeltentamen 2	3
Responsible Instructor	Dr. I.M. Smit	
Contact Hours / Week x/x/x/x	See AESB1211	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	See AESB1211	
Study Goals	See AESB1211	
Education Method	See AESB1211	
Assessment	See AESB1211	
Expected prior Knowledge	See AESB1211	
Academic Skills	See AESB1211	
Literature & Study Materials	See AESB1211	
Judgement	See AESB1211	
Permitted Materials during Exam	See AESB1211	
Collegerama	No	

AESB1211 Toets 4	*Deeltoets periode 2*	*0*
Responsible Instructor	Dr. I.M. Smit	
Contact Hours / Week x/x/x/x	See AESB1211	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	See AESB1211	
Study Goals	See AESB1211	
Education Method	See AESB1211	
Assessment	See AESB1211	
Expected prior Knowledge	See AESB1211	
Academic Skills	See AESB1211	
Literature & Study Materials	See AESB1211	
Judgement	See AESB1211	
Permitted Materials during Exam	See AESB1211	
Collegerama	No	

AESB1213	**Probability and Statistics**	**5**
Responsible Instructor	N. de Kleijn	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	4 h/w lecture, 2 h/w exercises, 2 h/w lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course covers the basics of probability theory and statistics. Topics in probability include discrete and continuous random variables, probability distributions, conditional probability and stochastic independence, joint distributions, sums and functions of random variables, expectation, variance and covariance, the law of large numbers, and the central limit theorem. Topics in statistics include random sample and sample distributions, graphical summaries, estimation and (un)biased estimators, confidence intervals, hypothesis testing and type I and II errors, the t-test.	
Study Goals	At the end of the course, the students will be able to: - Compute (conditional) probabilities; - Perform computations with discrete and continuous random variables and apply the law of large numbers and the central limit theorem; - Identify and use numerical values that summarize the information contained in random variables; - Relate random variables to statistical experiments and select statistical models suitable for describing real-life situations; - Analyze data by means of graphical and numerical summaries; - Define and evaluate statistical procedures for conducting statistical inference (estimation and hypothesis testing).	
Education Method	Lectures (4 h/w), exercise classe (2 h/w) and labs (2 h/w)	
Assessment	Two exams: midterm about first part of the course (probability theory) and a final exam at the end of quarter 3 about the second part (statistics). The exams will be partly multiple choice, partly open questions. Both exams count for 50% of the final grade (for each exam need at least 5.0, and average needs to be at least 5.8 to pass the course). In addition, a PASS for the lab assignments is required.	
Expected prior Knowledge	n.a.	
Academic Skills	-	
Literature & Study Materials	A Modern Introduction to Probability and Statistics - Understanding Why and How. Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. Springer Texts in Statistics, 2005, XVI, 488 p. 120 illus., Hardcover, ISBN: 1-85233-896-2	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Formula sheet (available on Brightspace, bring your own copy) and non-graphical calculator.	
Collegerama	No	

AESB1213 Toets 1	*Lab*	*0*
Responsible Instructor	N. de Kleijn	
Contact Hours / Week x/x/x/x	4 h/w lecture, 2 h/w exercises, 2 h/w lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	See AESB1213	
Study Goals	See AESB1213	
Education Method	See AESB1213	
Assessment	See AESB1213	
Expected prior Knowledge	See AESB1213	
Academic Skills	See AESB1213	
Literature & Study Materials	See AESB1213	
Judgement	See AESB1213	
Permitted Materials during Exam	See AESB1213	
Collegerama	Yes	

AESB1213 Toets 2	*Midterm*	*2.5*
Responsible Instructor	N. de Kleijn	
Contact Hours / Week x/x/x/x	4 h/w lecture, 2 h/w exercises, 2 h/w lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	See AESB1213	
Study Goals	See AESB1213	
Education Method	See AESB1213	
Assessment	See AESB1213	
Expected prior Knowledge	See AESB1213	
Academic Skills	See AESB1213	
Literature & Study Materials	See AESB1213	
Judgement	See AESB1213	
Permitted Materials during Exam	See AESB1213	
Collegerama	No	

AESB1213 Toets 3	*Final Exam*	*2.5*
Responsible Instructor	N. de Kleijn	
Contact Hours / Week x/x/x/x	4 h/w lecture, 2 h/w exercises, 2 h/w lab	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	See AESB1213	
Study Goals	See AESB1213	
Education Method	See AESB1213	
Assessment	See AESB1213	
Expected prior Knowledge	See AESB1213	
Academic Skills	See AESB1213	
Literature & Study Materials	See AESB1213	
Judgement	See AESB1213	
Permitted Materials during Exam	See AESB1213	
Collegerama	No	

AESB1230	Geology 2: North West Europe	5
Responsible Instructor	Drs. J.C. Blom	
Contact Hours / Week x/x/x/x	0/0/7/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Geology I	
Course Contents	Minerals and Rocks, Geology of NW Europe, Structural geology.	
Study Goals	The goal of the Geology 2 course is to provide students with knowledge and skills to understand the processes over time that lead to the formation of North West Europe, as well as the identification of some minerals and rocks that can be found in the subsurface of North West Europe and elsewhere	
Study Goals continuation	Knowledge: Students will be able to: - Explain the history and structure of the subsurface of North West Europe, as well as the natural processes that led to the formation of the present day North West Europe, with special emphasis on the Netherlands. - Describe the origin, formation and location of the different types of geologic resources found in North West Europe. - Explain the processes that lead to the formation of rocks (igneous, metamorphic and sedimentary). - Describe rock characteristics (texture, mineralogy) and use this to decide on rock names. - Demonstrate connections between rock forming processes and rocks (rock cycle). - Describe the tectonic processes and different deformational styles that played a role in the formation North West Europe, as well as other regions. - Explain the processes that play a role in the deformation of the Earth's crust. Skills: Students will be able to: - macroscopically identify 20 different rock forming minerals - Analyze the texture and mineralogy of rocks (igneous, metamorphic and sedimentary) macroscopically and subsequently able to infer the way they formed and to identify the rocks. - Draw cross sections illustrating the subsurface of North West Europe	
Education Method	The course Geology 2 is a combination of self study, lectures, laboratory work with practical experience on the macroscopic rock determination, a practical with maps and report writing. The Geology 2 course is composed of three parts dedicated to: 1) general knowledge and practical skills needed for the recognition of different minarals and rock types; 2) general knowledge of the Geology of North West Europe; 3) general knowledge and practical skills in Structural Geology. These last two parts will be taught in an integrated fashion, with a combination of self study, the writing of reports, and feedback during the contact hours.	
Literature and Study Materials	Part 1 will use a reader and handouts from Brightspace. For parts 2 and 3 handouts of lectures will be made available on Brightspace. In addition material from "Geology of the Netherlands", which will be available through Brightspace, and emodules from "Structural Geology", by Haakon Fossen (Cambridge University Press) will be used during class.	
Books	none mandatory, although Structural Geology by Haakon Fossen may be helpful.	
Reader	all information will be made available on Brightspace in the form of powerpoints and links to websites.	
Prerequisites	Geology 1	
Assessment	All three parts of the course will be tested with a written exam. Judgement: All three blocks count equally for the final grade. --- UPDATE dd 2-2-2021 due to the COVID-19 measures --- The course will be assessed with three online tests.	
Permitted Materials during Tests	None	
Remarks	The General Geology is valid for 5 EC which corresponds to 140 hrs, that is, 53 contact hours and 87 hrs of self-study. The 53 hours will be organized as follows: 36 hrs of lectures/feedback sessions (pt.1:12; pt.2: 12; pt.3: 12) 12 hrs of practical work (pt. 1: 12;) 5 hours: Three tests will be held at the end of each block The self-study hours will be divided as follows: About 21 hours of preparation for intermediate tests (7 per test), about 66 hours for preparation/working out lecture hours and practicals.	
Contact	Drs. J.C Blom J.C.Blom@tudelft.nl tel: +31 (0)15 27 83628	
Expected prior Knowledge	AESB 1130, Geology 1	
Academic Skills	For the Structural and NWE part, students will be asked to prepare before the scheduled contact hours, so these hours can be used to ask questions regarding the parts that were not understood. In addition to this, students will be asked to perform a weekly research project on the geology of part of NW Europe, and present the results of this research in a report, on which they will receive feedback.	
Literature & Study Materials		
Judgement	All three blocks count equally for the final grade. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	none	
Collegerama	No	

AESB1230 Toets 1	North West Europe	0
Responsible Instructor	Drs. J.C. Blom	
Contact Hours / Week x/x/x/x	Refer to AESB1230	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Refer to AESB1230	
Study Goals	Refer to AESB1230	
Education Method	Refer to AESB1230	
Assessment	Refer to AESB1230	
Expected prior Knowledge	Refer to AESB1230	
Academic Skills	Refer to AESB1230	
Literature & Study Materials	Refer to AESB1230	
Judgement	Refer to AESB1230	
Permitted Materials during Exam	Refer to AESB1230	
Collegerama	No	

AESB1230 Toets 2	Rock Types	0
Responsible Instructor	Drs. J.C. Blom	
Contact Hours / Week x/x/x/x	Refer to AESB1230	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Refer to AESB1230	
Study Goals	Refer to AESB1230	
Education Method	Refer to AESB1230	
Assessment	Refer to AESB1230	
Expected prior Knowledge	Refer to AESB1230	
Academic Skills	Refer to AESB1230	
Literature & Study Materials	Refer to AESB1230	
Judgement	Refer to AESB1230	
Permitted Materials during Exam	Refer to AESB1230	
Collegerama	No	

AESB1230 Toets 3	Structural Geology	0
Responsible Instructor	Drs. J.C. Blom	
Contact Hours / Week x/x/x/x	Refer to AESB1230	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Refer to AESB1230	
Study Goals	Refer to AESB1230	
Education Method	Refer to AESB1230	
Assessment	Refer to AESB1230	
Expected prior Knowledge	Refer to AESB1230	
Academic Skills	Refer to AESB1230	
Literature & Study Materials	Refer to AESB1230	
Judgement	Refer to AESB1230	
Permitted Materials during Exam	Refer to AESB1230	
Collegerama	No	

AESB1243	Grand Challenges and Applied Earth Sciences	9

Responsible Instructor	Dr. A. Barnhoorn
Instructor	Dr.ir. D.J.M. Ngan-Tillard
Contact Hours / Week x/x/x/x	8/8/0/0
Education Period	1 2
Start Education	1
Exam Period	2 3
Course Language	English
Course Contents	This first year introduction course of the BSc Applied Earth Science will provide the students an introduction and overview of the Future Challenges in the field of Applied Earth Sciences. The students will survey and link the Global Sustainable Development Challenges of the United Nations to the field of Applied Earth sciences (topics and disciplines). In the first half of the course (Q1) the focus is on working on a report on a specific grand challenge topic within the field of Applied Earth Science. The second half of the course (Q2) the students are introduced weekly to the different disciplines of applied earth science (Earth Resources, Remote Sensing, Climate Change, Material Properties, Mechanics, Heat, Flow and Environmental impact). With this different discplines, the student obtains a good overview of the topics that they focus on in their BSc degree. The student will obtain writing and presentation skills, group work, basic Python and emphasis will be placed on critical thinking / argumentation / posing right questions, planning, study and work attitude throughout the course and set goals for their future op life, interests and goals.
Study Goals	Knowledge Students should be able to: Explain what the field of Applied Earth Sciences studies and its relevance for the future; Understand the context in which an Applied Earth Science engineer operates; Explain the structure of the AES BSc programme in the context of geology, raw materials, societal development, construction. Show the role of AES in creating grand challenges/ sustainability goals of the UN; Show the role of AES in contributing to a solution for grand challenges/ sustainability goals of the UN; Apply geological, measuring, engineering, material properties, flow and data knowledge to solving a grand challenge in Applied Earth Sciences. Able to apply basic coding in Python (incl. use of loops, functions, numpy, matplotlib, numpy, pandas) Attitude Students should be able to: Take and active role in a group project; Take responsibility for their part of the group work, preparation of classes and living up to their own study planning; Use feedback from tutors and lectures to improve their work; Critically analyse, value and summarize results from literature and practicals. Reflect on their future op life, interests and goals
Education Method	This course contains of a mix of lectures, practicals, exercise, self study and group work hours.
Assessment	Group Final Advisory report, oral presentation, written exam, hand-in assignments (e.g. Python, goal setting)
Expected prior Knowledge	n.a.
Academic Skills	Skills: Students should be able to: Successfully work in groups; Perform data processing + visualisation and presentation of data using Python; Formulate a for Applied Earth Sciences engineer relevant research question on Grand Challenges; Collect, analyze, and summarize relevant technical/scientific literature with guidance; Write as a team an advisory report which conforms to the guidelines of a advisory report, with guidance; give an oral presentation of 10 to 15 minutes which conforms to the guidelines for presentation. Contribute (challenging, criticizing, explaining and defending statements) to a societal debate; Develop a self-evaluation strategy for problem solving (via continuously posing questions); Develop a Strategy to efficiently read/learn/filter and combine/ synthesize information. Reflect on their future op life, interests and goals
Literature & Study Materials	All material will be provided as handouts, readers, presentation slides via Brighspace throughout the course.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Group final advisory report Grand Challenge (50%) Presentation Skills (50%) Written Exam (35%) All three grades =>5, combined grade =>5.8 Hand-in assignments for Python and Goal Setting (Pass/Fail)
Permitted Materials during Exam	Pencil, Pen, non-graphical calculator

AESB1243 Toets 1	*Grand Challenges Advisory Report*	*4.5*

AESB1243 Toets 2	*Grand Challenges Presentation*	*1.35*

AESB1243 Toets 3	*Grand Challenges Exam*	*3.15*

AESB1320-17	Mechanics	5
Responsible Instructor	Dr. R.E.M. Riva	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Dr.ir. B. Wouters	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Lectures + analytical exercises. Chapters 2 - 14 Essential University Physics, Volume 1, Richard Wolfson. Subjects: motion in a straight line; motion in two and three dimensions; force and motion; using Newton's laws; work, energy and power; conservation of energy; gravity; system of particles; rotational motion; rotational vectors and angular momentum; static equilibrium; oscillatory motion; wave motion. The acoustic wave equation, stress and strain (hand-outs).	
Study Goals	Students will be able to understand and to apply basic principles of classical mechanics (including motion in two and three dimensions, forces, Newton's laws, friction, work, energy, conservation of energy, linear momentum, gravity, center of mass, rotational motion, angular momentum, static equilibrium, oscillatory motion, mechanical waves). At the end of this course, students will be able to solve classical mechanics problems (including those about Newton's laws, conservation of energy, linear and angular momentum, gravity, rotational motion, static equilibrium, oscillatory motion, mechanical waves) of difficulty comparable to the difficulty of the problems in the lecture book. In addition, students will be able to describe the acoustic wave equation as used in seismic applications, as well as the concepts of stress and strain.	
Education Method	A combination of lectures and analytical exercises	
Course Relations	Fundamental course related to all follow up courses	
Assessment	Written Exam (100%). For first-year students, admission to the written exam at the end of the lecture period requires completion of the two classroom tests (not graded).	
Remarks	Lectures: 12x2 =24 hrs Exercises: 8x2=16 hrs Classroom tests: 2x2=4 hours Homework assignments: 6x3=18 hrs Self-study: 75 hrs Exam: 1x3=3 hrs Total: 140 hours	
Expected prior Knowledge	Mathematics 1, Mathematics 2	
Academic Skills	Critical thinking. Creative use of knowledge to solve new problems. Capacity to work with abstract concepts. Planning and responsibility (timely study crucial to pass exam).	
Literature & Study Materials	Book: Essential University Physics, Volume 1, by Richard Wolfson, published by Pearson. Hand-outs over the acoustic wave equation.	
Judgement	100% written exam (paper based) at the end of the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Open-book exam: use of textbook, hand-outs and lecture slides is allowed. Any other material is not allowed.	
Collegerama	Yes	

AESB1411	Linear Algebra	5
Responsible Instructor	Dr. E. Emsiz	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	- Solving systems of linear equations (using Gaussian elimination) - Vectors in R^n - Linear dependence/independence - Matrix algebra (sum, product, inverse) - LU decomposition of a matrix - Linear transformations - Subspaces in R^n (basis, dimension); column space and null space - Determinants - Eigenvalues and eigenvectors - Diagonalization - Inner products and orthogonal sets - Orthogonal projections - Method of least squares and linear models - Symmetric matrices	
Study Goals	An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample. After successfully completing the course you will be able to: - Solve systems of linear equations. - Identify if a system of linear equations is solvable, and if the solution is unique. - Show the equivalence between a vector equation, a matrix equation and a system of linear equations. - Give a geometric description of solutions sets of vector equations. - Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence. - Identify if a transformation is linear and determine the standard matrix of a linear transformation. - Determine the matrix of geometric linear transformations. - Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties. - Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics. - Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace. - Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem. - Determine the LU-decomposition of a matrix. - Determine the determinant of a matrix by applying the properties of a determinant. - Apply the properties of determinants (product, transpose, inverse). - Determine the eigenvalues and eigenvectors of a matrix. - Determine the algebraic and geometric multiplicity of a matrix. - Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix. - Analyze the case of complex eigenvalues and eigenvectors. - Determine the matrix of a linear transformation with respect to a basis. - Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets. - Calculate the orthogonal projection on a subspace. - Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process. - Explain the normal equation of the least-squares method and apply the least-squares method to linear models. - Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices.	
Education Method	Online lectures blended with on-campus exercise sessions and (book and online) homework exercises. Disclaimer: information may change depending on the developments around the coronavirus.	
Assessment	Examination: digitally via assessment software with a written component if technically possible, otherwise completely on paper. P1: grade for first partial exam P2: grade for second partial exam You pass the course if the average of P1 and P2 is larger or equal to 5.8 (and P1 and P2 are larger or equal to 5.0).	
Contact	Dr. E. Emsiz	
Expected prior Knowledge	None.	
Academic Skills	- Interpretation - Analysis - Critical thinking - Consistent argumentation - Being master of the mathematical operations	
Literature & Study Materials	Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E. - Slides on Brightspace. - Prelecture videos. - Online homework assignments. - Book exercises - Old exams on Brightspace. Disclaimer: information may change depending on the developments around the coronavirus.	
Judgement	Examination: digitally via assessment software with a written component if technically possible, otherwise completely on paper. P1: grade for first partial exam P2: grade for second partial exam You pass the course if the average of P1 and P2 is larger or equal to 5.8 (and P1 and P2 are larger or equal to 5.0).	

Permitted Materials during Exam	None

AESB1420-17	**Electricity and Magnetism**	**5**
Responsible Instructor	Prof.dr.ir. H.W.J. Russchenberg	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Electric Charge, Force, and Field, Gauss's Law, Electric Potential, Electrostatic Energy and Capacitors, Electric Current, Magnetism, Electromagnetic Induction, Maxwell's Equations.	
Study Goals	Knowledge 1. Students are able to describe electric and magnetic fields in vacuum and matter and give a physical interpretation of bound and free charges, forces and energy. 2. Students are able to give physical interpretations of basic laws and relate them to static field conditions. 3. Students are able to describe the physics of Maxwells equations and reduce them to the equations for electromagnetic induction and to static equations. Skills 5. Students are able to solve electrostatic and electrodynamic equations for simple measurement configurations as used in applied earth sciences.	
Education Method	Just-in-time teaching, active work forms, self study. Students are expected to study the material before coming to class, only difficult concepts are taught in class, students are challenged to discuss and answer questions in class, ask and receive feedback from instructors.	
Assessment	Written test in week 11.	
Expected prior Knowledge	Preceding courses Mathematics and Mechanics of Applied Earth Sciences, first year	
Academic Skills	analytical, abstract, conceptual thinking, problem solving	
Literature & Study Materials	Essential University Physics, volume 3, second edition; author: R. Wolfson; chapters 1-5, 7, 8, 10 (or 20-24, 26,27, 29)	
Judgement	Written exam. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	none	
Collegerama	Yes	

AESB1430-14	Geology 3: geological systems and excursion	5
Responsible Instructor	Dr. G. Bertotti	
Instructor	Drs. J.C. Blom	
Instructor	Dr. J.E.A. Storms	
Contact Hours / Week x/x/x/x	0/0/0/7	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Geology 1: basics Geology 2: North West Europe	
Course Contents	The General Geology III is composed of two main parts. Part 1 consists of a 5 days excursion led by staff members of the Department. Geological phenomena are studied in the Belgian Ardennes, Luxembourg and the German Eifel during a 5 days excursion. The geological issues are (partly) discussed during the geological lectures and practicals prior to the trip Topics in the Belgian Ardennes: Sedimentary rocks (conglomerate, sandstone, shale, limestone and their fossil content), Reservoir properties of these rocks, the Caledonian and Hercynian deformation of rocks, various structures: layering, foliation, fractures, folds, unconformities, the weathering and erosion of rocks, stability problems within the rock units and geological engineering solutions Topics Luxembourg: Study of Jurassic iron oolites Topics in the German Eifel: Study of various types of volcanoes and associated volcanic rocks and deposits, Reconstruction of the volcanic history of the Eifel. Outcrops and structures visited during the excursion might change Part 2 of the course (Prof. G. Bertotti) aims at integrating all the knowledge and experiences gathered by students during previous geological courses and during the excursion and use them to address a range of fundamental and applied geological topics. Particular attention will be devoted to the genesis and evolution of sedimentary basins. Part 2 is organized in two components, one providing a summary of relevant geological notions, the second one is composed of two assignments during which students will work on real-world case studies tackling theoretical and practical issues.	
Study Goals	The overarching goal of the General Geology III course is to provide students basic knowledge and practical skills to understand and predict the functioning of geological systems.	
Study Goals continuation	Knowledge: Students will be able to describe: - The main tools used to reconstruct the geological record from different kinds of data. This includes information from the field. - Tectono-sedimentary processes active in extensional domains inclusive of their applied geology relevance. - Tectono-sedimentary processes active in contractional domains inclusive of their applied geology relevance. Skills: After the excursion, students are able to: - carry out measurements with a geological compass - analyze sedimentary rocks and structures in the field and is able to reconstruct the depositional environments. - to describe and analyze deformation structures and is able to reconstruct the tectonic history of the area. - to describe and analyze volcanic rocks and volcanic (also pyroclastic) deposits and is able to reconstruct the volcanic history of the area. - Integrate a variety of techniques (acquired during the first two modules) to understand specific geological settings. - Describe and interpret the geological record accessed during the excursion and provide answer to specific and general questions derived from the excursion topics. after the 2nd part students will be able to - perform a first general interpretation of seismic lines - extract the main sedimentological and structural properties - derive relevant information on relevant applied geological issues such as hydrogeology, hydrocarbon geology etc (depending on the nature of the assignments)	
Education Method	The course Geology 3 is a combination of classes and laboratory work with practical experience and excursion.	
Course Relations	This course is is key to follow other geological courses in the 2nd year	
Literature and Study Materials	The book adopted for the Geology 3 class is Understanding Earth (Grotzinger and Jordan). Handouts of classes will be available on Brightspace. The excursion is supported by an excursion guide.	
Prerequisites	Participation to the excursion is only allowed after succesful completion of Geology 1 (AESB1130) (final mark of at least 5.8)	
Assessment	Part 1. A report has to be written during the excursion. This report (prepared by two persons) must be submitted before arrival in Delft. In the excursion guide items are outlined per stop. The report must discuss these items; the items serve as a guideline for the judgment. Part 2 will be assessed on the basis of the results of the reports of the two practicals and of a written exam Judgement: Part 1 and part 2 will count each for 50% of the total mark. Part 1: The judgment of the report is based on the items mentioned above. The grade should be at least 6.0. An insufficient grade can be upgraded to a 6.0 by writing a literature review on an excursion-related subject. This literature review takes about 5 days to complete. If the literature review is insufficient for raising the grade to 6.0 there is one last chance to improve the mark by improving the literature review. After failing again the excursion has to be followed again; but now at ones own expense. For part 2, the two reports will count for 25% each, the remaining 50% being awarded on the basis of the written exam. The grades of the reports must be 5 or higher	
Permitted Materials during Tests	Coloured pencils and pocket calculator. Further specifics will be given.	
Contact	Prof. Dr. G. Bertotti G.bertotti@tudelft.nl tel: +31 (0)15 27 86033	
Expected prior Knowledge	General geology 1 and 2	
Academic Skills	Master the knowledge acquired in previous courses and ability to apply it to understand outcrops and perform the practicals	
Literature & Study Materials	Handouts, material on Brightspace and "Understanding Earth"	
Judgement	Part 1 and part 2 will count each for 50% of the total mark. Part 1: The judgment of the report is based on the items mentioned above. The grade should be at least 6.0. An insufficient grade	

	can be upgraded to a 6.0 by writing a literature review on an excursion-related subject. This literature review takes about 5 days to complete. If the literature review is insufficient for raising the grade to 6.0 there is one last chance to improve the mark by improving the literature review. After failing again the excursion has to be followed again; but now at ones own expense. For part 2, the two reports will count for 25% each, the remaining 50% being awarded on the basis of the written exam. The grades of the reports must be 5 or higher For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	colored pencils, ruler, protractor
Collegerama	No

AESB1430-14 Toets 1	*Report excursion*	*0*
Responsible Instructor	Dr. G. Bertotti	
Contact Hours / Week x/x/x/x	NA	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Collegerama	No	

AESB1430-14 Toets 2	*Geological systems*	*0*
Responsible Instructor	Dr. G. Bertotti	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Assessment	The achievement of the goals of the course and, more specifically of its practical part will be assessed by evaluation of the individual practical reports (60%) and by the exam (40%) which will take place with the "open book" mode.	

AESB1440-21	Methodology of Geophysics and Remote Sensing	5
Responsible Instructor	Dr. C. Weemstra	
Instructor	Dr. P.G. Ditmar	
Instructor	Dr.ir. G.G. Drijkoningen	
Instructor	Dr. M.P.M. Franssen	
Instructor	W.B. Vermaat	
Contact Hours / Week x/x/x/x	x/x/x/5 Applied Geophysics & Remote Sensing Methodology: 20 hours Integrated project: 20 hours Philosophy of Science lectures by TBM: 5 hours	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Students are given a broad overview of Applied-Geophysics and Remote-Sensing methods. In addition, the general, scientific basis underlying such methodology is introduced (e.g., (i) the interplay between data, models and theory and (ii) the concept of resolution). The course includes the main concepts of scientific methodology via some separate lectures on `philosophy of science'. The methods are studied in a societal and/or economically relevant context, and some of them are applied in a field experiment. In groups, the students will work on three large assignments (problem-based learning): 1. An in-depth study of a specific applied-geophysics method (e.g., refraction/reflection seismics, electric resistivity imaging, etc.) in the context of a societally and/or economically relevant problem. The groups are expected to find resources themselves, which will train them to find and value applied-geophysics literature, and independently acquire relevant knowledge. This assignment commences with a general introduction (lecture) in applied-geophysics methodologies. To monitor the group dynamics, ensure progress and focus, and provide feedback, several consults are scheduled over the course of the assignment. A draf report is submitted which is also provided with feedback, enabling students to shape their writing skills. At the end a final report is submitted and the results are presented orally. 2. An in-depth study of a specific remote-sensing method (e.g., gravimetry, hyperspectral imaging, etc.) in the context of a societally and/or economically relevant problem. The groups are expected to find resources themselves, which will train them to find and value remote-sensing literature, and independently acquire relevant knowledge. This assignment commences with a general introduction (lecture) in remote-sensing methodologies. To monitor the group dynamics, ensure progress and focus, and provide feedback, several consults are scheduled over the course of the assignment. A draf report is submitted which is also provided with feedback, enabling students to shape their writing skills. At the end a final report is submitted and the results are presented orally. 3. Field application of some of the applied-geophysics and remote-sensing methods studied in assignments 1 and 2. For applied geophysics, often the electric resitivity method, low-frequency EM or (shear-wave) seismic method is taken, while precise positioning with GPS/GNSS will be taken as the Remote-Sensing method. This assignment includes some literature study (bibliographical study), defining goals and hypotheses, modelling of the experiments, planning of the field experiments, carrying out the field experiments, processing of the field data, and interpretation of the data and results. The results are subsequently compared against the initial hypothesis (modelling results), and conclusions are drawn in terms of the initial goals (geological/environmental in nature). The entire project (workflow) and its results are presented orally (after the lecturers have given feedback on the draft presentation). Importantly, the oral presentations at the end of the assignments will familiarise all students with the different applied-geophysics and remote-sensing methods (and hence not only with the methodology they studied with their group).	
Study Goals	Knowledge: Students will be able to - apply fundamental knowledge (mathematics, physics, geology) in some elementary Applied-Geophysics and Remote-Sensing projects. - relate basic physical laws (on mechanics and electricity & magnetism) to applied-geophysics and remote-sensing methods. - classify different Applied-Geophysics and Remote-Sensing methods. - link theoretical concepts, resulting from basic physical laws, with practices in the field. - use an applied-geophysics and remote-sensing method to address a societal, scientifically, or economically relevant problem. Skills: students will be able to - simulate the result of field experiments using forward modelling techniques. - make a field-acquisition plan, acquire, process and interpret data obtained with an applied-geophysical and remote-sensing method. Soft skills: students will be able to: - approach a problem from a scientific point of view, using basic physical laws, observations/facts and models (philosophy of science). - write a scientific report. - give an oral presentation of a scientific subject. - retrieve and use relevant literature from the library for its use in a geoscience project. Attitude: students will be able to - work efficiently in groups not of ones own choice; to work together and delegate tasks; to be accountable to their group; and to deal with limitations of fellow group members.	
Education Method	Problem-based approach. Students will work in groups of 3 to 4. They will give oral presentations, carry out library/literature studies and write reports. They will apply knowledge/skills from philosophy of science to problems/projects. There will be a few introductory lectures about applied-geophysics and remote-sensing methodologies to support the problem-based education. The separate consults with the groups, and the (formative) feedback on the drafts of the presentations and reports, allow the students to meet the learning objectives (or study goals).	
Course Relations	This course has a relation to the courses: - Geophysical Methods for Subsurface Characterization (2nd year) - Instrumentation & Signals (2nd year) - Geological Fieldwork & Remote Sensing (2nd year) - Field Exploration & Exploitation (3rd year) - Applied-Geophysics and Remote-Sensing courses in the MSc program	
Assessment	- Applied-geophysics and remote-sensing methods (assignments 1 & 2): written report (both assignments) plus an oral presentation by the group (to all fellow students) of one of the two assignments.	

	- Integrated field project (assignment 3): oral presentation (to all fellow students) by the group. - Separate student-individual summative assessment is performed by a lecturer of TBM (including formative feedback). Consequently, each student in a group has to give at least one oral presentation. - Philosophy of Science is assessed separately. The final grade is then built up as follows: - Applied-geophysics and remote-sensing methodology (assignments 1 & 2): 60% (Register for AESB1440-21 Exam 01) - Integrated field project (assignment 3): 30% (Register for AESB1440-21 Exam 02) - Presentation skills: 5% (Register for AESB1440-21 Exam 03) - Philosophy of Science: 5% (Register for AESB1440-21 Exam 04)
Special Information	The total study load is 5 EC. This is divided as follows (see also assessment): Applied-geophysics and remote-sensing methodology: 3.0 EC (Exam 01) Integrated project (including self-acquired field data): 1.5 EC (Exam 02) Presentation skills (soft skills): 0.25 EC (Exam 03) Philosophy of Science: 0.25 EC (Exam 04)
Contact	Cornelis Weemstra C.Weemstra@tudelft.nl
Expected prior Knowledge	Since this is a course in which a large number of geophysical and remote sensing methods are introduced, almost all courses in the first year of the AES pogramme are relevant. The following, however, are particularly important: - Grand challenges and applied Earth sciences (for the report writing and presentation skills) - Mathematics 1 - Mechanics - Electricity and magnetism
Academic Skills	Scientific skills: approach a problem from a scientific point of view (Philosophy of Science) Soft skills: report writing (assessed), oral presentation (assessed) and library skills (feedback)
Literature & Study Materials	Acquiring relevant knowledge is part of the learning objectives. Of course, the students will be supported by the lecturers for this. When necessary, relevant articles and information will be made available via Brightspace.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	(no exam: only reports and presentations)
Collegerama	No

AESB1440-21 Exam 01	***Final exam / Lab / Midterm***	***3***
Responsible Instructor	Dr. C. Weemstra	

AESB1440-21 Exam 02	***Final exam / Lab / Midterm***	***1.5***
Responsible Instructor	Dr. C. Weemstra	

AESB1440-21 Exam 03	***Final exam / Lab / Midterm***	***.25***
Responsible Instructor	Dr. C. Weemstra	

AESB1440-21 Exam 04	***Final exam / Lab / Midterm***	***.25***
Responsible Instructor	Dr. C. Weemstra	

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AESB2110-21	Mathematics 4: Differential Equations	5
Responsible Instructor	Dr.ir. M. Keijzer	
Module Manager	Ir. D. den Ouden-van der Horst	
Contact Hours / Week x/x/x/x	4/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	After a short recap, more on differential equations: the Laplace transform, systems of differential equations and partial differential equations.	
Study Goals	After completing this course, the student should be able to - classify differential equations, tell their appropriate number of conditions, and name methods to solve the different types. - calculate the general solutions for linear, first and second-order, homogeneous and inhomogeneous, ordinary differential equations with constant coefficients. - rewrite a solution in the form A*cos(omega t) + B*sin(omega t) as C*cos(omega t + phi). - use the Laplace transform to solve linear initial value problems that may contain terms with a step function or a Dirac delta function. - rewrite a higher-order differential equation as a system of first-order differential equations. - solve a system of linear, first-order differential equations with constant coefficients and determine particular solutions with the method of undetermined coefficients. - sketch a phase portrait for a system of two almost linear, autonomous differential equations and determine the types of the equilibrium points. - construct ordinary differential equations for some basic physical processes and show some understanding of the relation between the differential equations and the physical processes that are modelled by them. - extend a function on an interval to an even or odd periodic function and express the result into a Fourier series. - apply the method of separation of variables to partial differential equations (the heat conduction equation, wave equation and Laplace's equation) with appropriate initial and boundary conditions. - demonstrate some understanding between the boundary value problems and the physical processes that are modelled by them.	
Education Method	Lectures, 2x2 hours a week Practical 6x2 hours Online course: Mathematical Modelling Basics	
Assessment	- The practical, consisting of six practical sessions. Judgement: The practical has to be completed, by finishing each of the six parts with a satisfactory percentage. The final result for the practical is a V (pass) or an O (fail). - The online course Mathematical Modelling Basics: a collection of exercises and tests have to be done. Judgement: The online course has to be completed, by finishing the exercises and tests in the course with a satisfactory percentage. The result is a V or an O. - A written exam with open questions. Judgement: The written summative exam is graded with a correction model. The number of points that can be obtained for each question is shown on the exam and is relative to the importance of the learning goal that is tested and to the time it takes to answer the question. The result is a grade 1-10, which serves as the final grade for the course.	
Expected prior Knowledge	Calculus Linear Algebra	
Academic Skills	Critical thinking when modelling practical problems and validating the solutions.	
Literature & Study Materials	- Book: W.E. Boyce & R.C. DiPrima: Elementary Differential Equations and Boundary Value Problems, global edition (11th) - Videos: One for each lecture, to prepare for the lecture - Preparatory exercises: Online Grasple exercises to prepare for the lecture - Lecture slides with multiple choice questions - DIY exercises: Online Grasple exercises with larger problems broken up in tiny steps with a lot of feedback - Homework exercises from the book to practice that are also available in Grasple, but without much feedback in order to simulate exam conditions. - Context videos: demonstrations of applications of differential equations	
Judgement	The practical has to be completed, by finishing each of the six parts with a satisfactory percentage. The result is a V (pass) or an O (fail). The online course has to be completed, by finishing the exercises and tests in the course with a satisfactory percentage. The result is a V or an O. The written summative exam is graded with a correction model. The number of points that can be obtained for each question is shown on the exam and is relative to the importance of the learning goal that is tested and to the time it takes to answer the question. The result is a grade 1-10, which serves as the final grade for the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	A formula sheet with Laplace transforms is handed out. Additionally for an on campus exam: a simple calculator Additionally for an online exam: the book Boyce & DiPrima, slides from the course, your own notes, a (graphical) calculator.	
Collegerama	No	

AESB2110-21 toets 1	Final exam	5
Responsible Instructor	Dr.ir. M. Keijzer	
Course Contents	See AESB2110-21	
Study Goals	See AESB2110-21	
Education Method	See AESB2110-21	
Assessment	See AESB2110-21	
Expected prior Knowledge	See AESB2110-21	
Academic Skills	See AESB2110-21	
Literature & Study Materials	See AESB2110-21	
Judgement	See AESB2110-21	
Permitted Materials during Exam	See AESB2110-21	
Collegerama	No	

AESB2110-21 toets 2	Lab: practical	0
Responsible Instructor	Dr.ir. M. Keijzer	
Course Contents	See AESB2110-21	
Study Goals	See AESB2110-21	
Education Method	See AESB2110-21	
Assessment	See AESB2110-21	
Expected prior Knowledge	See AESB2110-21	
Academic Skills	See AESB2110-21	
Literature & Study Materials	See AESB2110-21	
Judgement	See AESB2110-21	
Permitted Materials during Exam	See AESB2110-21	
Collegerama	No	

AESB2110-21 toets 3	Lab: MOOC	0
Responsible Instructor	Dr.ir. M. Keijzer	
Course Contents	See AESB2110-21	
Study Goals	See AESB2110-21	
Education Method	See AESB2110-21	
Assessment	See AESB2110-21	
Expected prior Knowledge	See AESB2110-21	
Academic Skills	See AESB2110-21	
Literature & Study Materials	See AESB2110-21	
Judgement	See AESB2110-21	
Permitted Materials during Exam	See AESB2110-21	
Collegerama	No	

AESB2122	Signals and Systems with Python	5
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	4 (lect.) (of which instruction in weeks 6 and 8, instead of lect.) and 2 instruction (computer practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides an introduction to signal theory and to how to model the relation between the input and output of systems. Topics covered: - Fundamental signals and operations with signals (scaling, shifting, etc). - Concepts of Energy and Power. - Frequency domain representation of signals and the transformation and relation between the time and the frequency domain. - Types of systems (causal vs non-causal, linear vs non-linear, time-invariant). - Characterization of Liniear Time Invariant (LTI) systems by their impulse and frequency response, and the relation between input and output in time and frequency domains. - Periodic signals and Fourier series. - Fourier transform and its application to signals and systems. - The Discrete Fourier Transform (DFT) and its applications to spectral analysis. - Basic filters. In addition, during this course students will be introduced to data analysis and signal-processing oriented programming using Python.	
Study Goals	At the end of this course, the student will be able to: 1. Identify, represent graphically and operate with some fundamental signals (e.g. Dirac-delta, sinc function, etc.) and mathematical operations (e.g. shifting, scaling, etc.). 2. Explain the concepts of energy and power, and calculate these for a given signal 3. Identify basic properties of systems and be able to classify them based on these properties. 4. Determine the frequency domain representation (mathematically and graphically) of signals using Fourier Series decomposition for periodic signals, and the Fourier transform for aperiodic ones. 5. Determine or calculate the impulse responses of LTI systems and the corresponding frequency responses. 6. Apply the impulse response and the frequency response of a system to determine the output of a signal given a time or frequency-domain input. 7. Analyze a signal in the time-and frequency domain, and design a filter to isolate the relevant part of the signal. 8. Apply the signal-theory concepts acquired in the course to analyze and solve real-life problems described in natural language. 9. Apply fundamental notions of sampling theory to design an adequate signal acquisition and analysis scheme. 10. Decide (and execute) the right strategy to analyze or solve a signal and systems problem, in particular by choosing the appropriate domain.	
Education Method	Signals and Systems theory is covered in in-class lectures with some in-class exercise and classical homework. The python/practical part is covered in practical sessions.	
Assessment	The course is assessed through two exams (midterm and final), both including Python assignments and written questions. The weight of the Python component in the grades is 40%. In addition, students have to complete several assignments and participate in practical sessions that are evaluated with a pass/no-pass grade. Judgement:The course will be assessed though two tests (midterm and final). Each test will have a theory part and a Python programming part. A minimum of 5 at each test is required to pass the course.	
Expected prior Knowledge	The course is largely self contained, but students are expected to have a math level consistent with that of a second year AESB student. In particular, students should be comfortable with complex numbers, sketching functions, and basic differential equations.	
Academic Skills		
Literature & Study Materials	The course follows parts of "Signals and Systems" by Ziemer (abbreviated version is available via MV).	
Judgement	The course will be assessed though two tests (midterm and final). Each test will have a theory part and a Python programming part. A minimum of 5 at each test is required to pass the course. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	The exams will be open book / open notes. A calculator is also allowed.	
Collegerama	No	

AESB2122 Toets 1	Midterm	2.5
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	4 (lect.) (of which instruction in weeks 6 and 8, instead of lect.) and 2 instruction (computer practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See AESB2122	
Study Goals	See AESB2122	
Education Method	See AESB2122	
Assessment	See AESB2122	
Expected prior Knowledge	See AESB2122	
Academic Skills	See AESB2122	
Literature & Study Materials	See AESB2122	
Judgement	See AESB2122	
Permitted Materials during Exam	See AESB2122	
Collegerama	No	

AESB2122 Toets 2	Final Exam	2.5
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	4 (lect.) (of which instruction in weeks 6 and 8, instead of lect.) and 2 instruction (computer practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See AESB2122	
Study Goals	See AESB2122	
Education Method	See AESB2122	
Assessment	See AESB2122	
Expected prior Knowledge	See AESB2122	
Academic Skills	See AESB2122	
Literature & Study Materials	See AESB2122	
Judgement	See AESB2122	
Permitted Materials during Exam	See AESB2122	
Collegerama	No	

AESB2122 Toets 3	Lab/practical sessions	0
Responsible Instructor	Dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	4 (lect.) (of which instruction in weeks 6 and 8, instead of lect.) and 2 instruction (computer practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	See AESB2122	
Study Goals	See AESB2122	
Education Method	See AESB2122	
Assessment	See AESB2122	
Expected prior Knowledge	See AESB2122	
Academic Skills	See AESB2122	
Literature & Study Materials	See AESB2122	
Judgement	See AESB2122	
Permitted Materials during Exam	See AESB2122	
Collegerama	No	

AESB2140-21	Geophysical Methods for Subsurface Characterization	5

Responsible Instructor Dr.ir. G.G. Drijkoningen

Instructor Dr. K.H.A.A. Wolf

Instructor Dr.ir. E.C. Slob

Contact Hours / Week x/x/x/x

The contact hours are divided as follows:
- Reflection seismology: 24 (12 hours lectures, 12 hours practical)
- Electric: 9 hours (5 hour lectures, 4 hours practical)
- Logging methods: 14 hours (6 hours lectures, 8 hours practical)

Education Period 1

Start Education 1

Exam Period 1

Course Language English

Course Contents

The course has three parts:
- reflection seismology
- electrical sounding
- borehole logging,

and:
- combines these parts in an integrated manner.

The reflection-seismology part deals with the most important aspects of reflection seismology: theory of seismic waves and aspects of data acqusition and data processing (CMP processing, velocity analysis, stacking, and migration) are dealt with. The reflection-seismology part are supplemented by a practical of 4 afternoons, where the students will see the most important data-processing steps via exercises (in Python).
The electric-resistivity part deals with the basic aspects of electric resistivity measurements: theory, acquisition, converting data to apparent resistivity, modelling, sensitivity analysis and inversion. The theory part is supplemented by one afternoon of practical work with computer exercises for modelling and inversion of DC resistivity data.
The borehole-logging part deals with the most important types of borehole logging data, focusing on acoustic and induction/resistivity logs. In the borehole logging exercises the use of gamma-ray, density and neutron logs for rock type determination are explained. Combined logs are used to interpret the rock type, and link the log data to acoustic, elastic/geomechanical, and electrical rock properties. The practicals cover the interpretation of the lithology from log data, the derivation of elastic rock properties and depth calibration of seismic data of the seismology part of the course.

Another part of the course consists of an Integrated Assignment in which the students will get some raw seismic, DC-apparent-resistivity and logging data, and they need to use these data to create images of the subsurface that characterize the subsurface in terms of lithologies and the possible fluids therein.

Study Goals

At the end of the course you will be able to interpret seismic, electric and borehole-logging data to characterize the subsurface from a physical perspective and combine these interpretations for a lithological characterization of the subsurface.

Knowledge, students are able to:
- Explain the measurement of raw seismic data, in terms of seismic wave propagation;
- Model seismic data from a kinematic point of view;
- Transform raw seismic data to a migrated seismic image, via a kinematic model of the wave-propagation process;
- Explain the concept of apparent resistivity for any array type and apply it to the response of two quarter spaces;
- Explain the relation between electric potential and electric current and explain which boundary conditions must be applied to current and potential when the conductivity shows a jump discontinuity;
- Compute current fractions in various depth ranges for a current electrode pair and explain 3D sensitivity of any electrode configuration;
- Interpret multiple log data and combine them to interpret the various lithologies and possible fluids in the subsurface and attribute rock physics characteristics to these lithologies and fluids;
- Use log data for the interpretation of seismic and DC-resistivity datasets.

Skills, students are able to:
- Process seismic data records to final seismic images for subsurface characterization;
- Invert apparent-resistivity data to a final resistivity image for subsurface characterization;
- Improve on their skills in Python;
- Improve on their academic writing skills.

Attitude, students have:
- Developed a critical attitude towards the limitations of their interpretations;
- Reflected on the complexity of using multiple datasets for improved subsurface characterization.

Education Method

As main teaching method, lectures will be given. For the practical exercises and the report of the final integrated exercise, the students work in groups of 2 behind the computer. The practicals are done under supervision in class, while for the final integrated exercise, the students are supposed to work on their own. The integrated exercise will consist of writing a report and obtain feedback from the teachers on this report before delivering the final report.

Course Relations

This course is related to the courses:
- Rock Mechanics & Rock Engineering (AESB2341),
- Soil Mechanics (AESB2330),
- Mechanics and Transport by Flow in Porous Media (AESB3340),
- Petrophysics & Image Analysis (AESB3341),
- Field Exploration Project (AESB3440),
- Geophysics courses in the MSc program,
- Log evaluation courses in the MSc program

Assessment

Formative tests: the practical exercises need to be finalized (no mark) before the final integration exercise can be done.

Summative tests: written exam and final report

The final grade will established via:
- Exam at the end of the course (50%)
- Report of the final integrated exercise (50%)

Pass mark is 6, each summative test must be passed by at least 5.0

Contact

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	Evert Slob e.c.slob@tudelft.nl tel: +31 15 278 8732
Expected prior Knowledge	Mechanics (AESB1320), Electricity & Magnetism (AESB1420), Signals & Systems with Python (AESB2122)
Academic Skills	Skills required: - Some knowledge of Python - Skills in academic writing.
Literature & Study Materials	Lecture notes and slides by G.G. Drijkoningen, E.C. Slob and K.H.A.A. Wolf. The lecture notes "Introduction to Reflection Seismology" can be ordered via the on-line service from the TU Delft; these lecture notes and all the other material can be freely downloaded as pdf from Brightspace.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Allowed at exam: - Pen and colour pencils - Non-graphical calculator - Ruler
Collegerama	No

AESB2140-21 Toets 1	***Final exam***	***2.5***
Responsible Instructor	Dr.ir. G.G. Drijkoningen	
Course Contents	(see description AESB2140-21)	
Study Goals	(see description AESB2140-21)	
Education Method	(see description AESB2140-21)	
Assessment	(see description AESB2140-21)	
Expected prior Knowledge	(see description AESB2140-21)	
Academic Skills	(see description AESB2140-21)	
Literature & Study Materials	(see description AESB2140-21)	
Judgement	(see description AESB2140-21)	
Permitted Materials during Exam	(see description AESB2140-21)	
Collegerama	No	

AESB2140-21 Toets 2	***Lab***	***2.5***
Responsible Instructor	Dr.ir. G.G. Drijkoningen	
Course Contents	(see description AESB2140-21)	
Study Goals	(see description AESB2140-21)	
Education Method	(see description AESB2140-21)	
Assessment	(see description AESB2140-21)	
Expected prior Knowledge	(see description AESB2140-21)	
Academic Skills	(see description AESB2140-21)	
Literature & Study Materials	(see description AESB2140-21)	
Judgement	(see description AESB2140-21)	
Permitted Materials during Exam	(see description AESB2140-21)	
Collegerama	No	

AESB2210-18	Mathematics 5	5

Responsible Instructor Ir. D. den Ouden-van der Horst

Contact Hours / Week x/x/x/x
4hrs lecture p/week
4hrs practical p/week

Education Period 2

Start Education 2

Exam Period
2
3

Course Language English

Course Contents
This is an introductory course in numerical analysis. Subjects of the course: introduction numerical analysis, interpolation, numerical differentiation, numerical methods for initial value problems, numerical methods for boundary value problems, numerical integration, methods to solve non-linear systems of equations.

Study Goals
At the end of this course you will be able to
1. interpolate between given data points using:
 o Lagrange interpolation polynomials;
 o first order splines;
 o cubic splines;
2. give upper bounds for the error for a given interpolation method, with and without the effect of rounding/measurement errors;
3. Approximate first and higher derivatives using, but not limited to
 o central differences;
 o forward differences;
 o backward differences;
4. derive a finite difference and show the order of the resulting finite difference;
5. estimate the error using Richardsons order and error estimation;
6. improve a finite difference using Richardsons extrapolation;
7. approximate the solution to nonlinear (systems of) algebraic equations using iterative methods such as
 o the Picard method;
 o the bisection method;
 o the Newton-Raphson method;
 o the Quasi-Newton method;
 o the Secant method;
 o the Regula-Falsi method;
8. select a suitable stopping criteria for a given method;
9. determine whether a given iterative method converges for a given problem and with which order;
10. derive a given iterative method;
11. use numerical integration to approximate integrals, more specifically, using
 o the (composite) left Rectangle rule;
 o the (composite) right Rectangle rule;
 o the (composite) Midpoint rule;
 o the (composite) Trapezoidal rule;
 o the (composite) Simpsons rule;
12. give upper bounds for the error for a given numerical integration method, with and without the effect of rounding/measurement errors;
13. solve initial value problems using numerical time-integration methods such as, but not limited to:
 o the Forward Euler method;
 o the Backward Euler method;
 o the Trapezoidal method;
 o the Modified Euler method;
 o the Fourth-order Runge-Kutta method;
14. derive the methods given above;
15. derive and analyse the local truncation error for a given numerical time-integration method, with and without use of the test equation;
16. determine or evaluate the stability for a given numerical time-integration method using the amplification factor;
17. choose between two or more given numerical time-integration methods, based on errors, stability and computational effort;
18. estimate the error using Richardsons order and error estimation;
19. solve boundary value problems using the finite difference method;
20. derive and analyse the local truncation error for a given finite difference scheme;
21. determine the stability for a given finite difference scheme using the Gershgorin Circle theorem;
22. choose between central or upwind finite differences;
23. combine a finite difference method with a numerical time-integration method to solve the transient heat equation in one spatial dimension;
24. derive and analyse the local truncation error for a given combination of a finite difference method and a numerical time-integration method;
25. determine the stability bounds for a given combination of a finite difference method and a numerical time-integration method;
26. analyse a given problem;
27. select appropriate numerical methods to solve a given problem;
28. design an algorithm to solve a given problem;
29. implement an algorithm to solve a given problem;
30. visualise the solutions of a given problem;
31. discuss the solutions of a given problem;
32. report on the solutions, the solution algorithm and the used numerical methods for a given problem.

Education Method This course consists of lectures and lab work.

Computer Use Lab work to be solved using Python.

Practical Guide
One Jupyter Notebook per assignment must be completed and submitted.

Notebooks can be submitted by a group of at most two students: these students must register as group in Brightspace (see above).

Only students in a registered group are allowed to cooperate on the assignment.

Notebooks can only be handed in through Assignments within Brightspace. Any notebook handed in any other way will not be assessed.

Distributing to others and/or receiving from others the assignment and/or the resulting notebook is prohibited. With others we mean any party not being your group partner, the lecturers and/or the assistants.

An action of one student counts as an action of the entire registered group.

If any of the above rules are violated by a group, any assessment of the Notebooks of the violating group will be revoked.

Assessment	The course is assessed by two component examinations: - a written exam (mark, rounded to tenths); - lab work practical containing 3 assignments with reports (V/O). There is also the possibility to do online exercises in the 2nd, which will result in a mark M, rounded to tenths. In the 2nd period the exam mark E and the mark M obtained in the same period give a final exam mark F using the formula F = max{E,0.85E+0.15M} if E>=5, rounded to tenths. Otherwise and/or in the 3rd period F = E. The lab work reports are graded as V or O, where the lab work result L is a V if all three reports are a V. If F>=5 and L=V, the final mark R of the course is R = F, rounded to halves and wholes. --- In case of exams not being able to take on-campus --- The following applies to the exams of AESB2210-18. The lab work of the course AESB2210-18 remains the same as in the study guide. The bonus system for the course remains the same as in the study guide. The exam will be as follows. There will be made use of online proctoring through RPNow. The exam questions will be done remotely through Grasple. During the exam only the use of the following is permitted: o Pen and paper o The built-in Grasple calculator Students have 90 minutes to enter answers in Grasple (105 if extra time is allowed). Within 15 minutes after submitting the answers in Grasple the students are allowed to submit his/hers handwritten elaborations in Brightspace. o Submission of handwritten elaborations is optional, so not mandatory. The assessment of the exam will be as follows: If the answer given in Grasple is correct, the student earns all of the (preset) total credits for the question. If the answer given in Grasple is incorrect, but stems from a predictable (programmable) mistake, you will earn a part of the (preset) total credits for that question. If the answer given in Grasple is incorrect, does not stem from a predictable (programmable) mistake and you have submitted handwritten elaborations in Brightspace within the set time frame, the handwritten elaboration for that question will be assessed. o This assessment will always result in less than all of the (preset) total credits for the question. If the answer given in Grasple is incorrect, does not stem from a predictable (programmable) mistake and you have not submitted handwritten elaborations in Brightspace within the set time frame, no credits will be given for that question.
Tags	Mathematics Numeric Methods Programming
Expected prior Knowledge	AESB1211, AESB1311, AESB2110-21, a Python course.
Academic Skills	Critical and analytical thinking, problem solving, writing a report (for the practical computer assignment)
Literature & Study Materials	Numerical Methods for Differential Equations (C.Vuik, F.J. Vermolen, M.B. van Gijzen, M.J. Vuik), Delft Academic Press, 2nd edition, 2015
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Non-programmable, non-graphical pocket calculator during on-campus exams
Collegerama	No

AESB2210-18 Toets 1	***Practical***	***2***
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Contact Hours / Week x/x/x/x	see description AESB2210-18	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	see description AESB2210-18	
Study Goals	see description AESB2210-18	
Education Method	see description AESB2210-18	
Assessment	see description AESB2210-18	
Expected prior Knowledge	see description AESB2210-18	
Academic Skills	see description AESB2210-18	
Literature & Study Materials	see description AESB2210-18	
Judgement	see description AESB2210-18	
Permitted Materials during Exam	see description AESB2210-18	
Collegerama	No	

AESB2210-18 Toets 2	*Exam*	*3*
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Contact Hours / Week x/x/x/x	see description AESB2210-18	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	see description AESB2210-18	
Study Goals	see description AESB2210-18	
Education Method	see description AESB2210-18	
Assessment	see description AESB2210-18	
Expected prior Knowledge	see description AESB2210-18	
Academic Skills	see description AESB2210-18	
Literature & Study Materials	see description AESB2210-18	
Judgement	see description AESB2210-18	
Permitted Materials during Exam	see description AESB2210-18	
Collegerama	No	

AESB2220-20	Chemical Thermodynamics	5
Responsible Instructor	Dr. A.P. Siebesma	
Contact Hours / Week x/x/x/x	4 hrs per week lectures (on-line) and 2 hrs per week exercises (on-site)	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Fundamental Concepts of Thermodynamics: - First Law of Thermodynamics : Energy, Heat & Work. Equation of state, Thermodynamical processes and cycles. - Second Law of Thermodynamics: Entropy, reversible vs irreversible processes. - Applications: Heat Engines, Heat Pumps, Mixing processes, Ideal Gas Mixtures and Humid Air, Phase Transitions, Pure Component Vapour-Liquid equilibrium, Thermodynamics of Mixtures. Multicomponent Phase Equilibria. Phase Diagrams. Chemical Reaction Equilibria.	
Study Goals	Knowledge: at the end of the course, the student is able to - to distinguish different forms of Energy - describe the difference between a ideal and non-ideal gas from a molecular perspective. - interpret the meaning of entropy - differentiate between reversible and irreversible processes. - evaluate thermodynamics properties and processes of materials in terms of the appropriate thermodynamics potentials. - Interpret phase diagrams of particular material. - Understand phase transitions and phase equilibria from a molecular perspective. Skills at the end of the course, the student is able to - Apply the first law of Thermodynamics to calculate energy transformations and changes in volume, pressure and temperature for reversible processes of an ideal gas. - Calculate efficiencies of energy cycles (engines, heat pumps). - Decide whether specified irreversible processes will take please spontaneously - Calculate entropy changes of irreversible processes (mixing processes, free expansion etc) - Calculate energy changes associated with phase changes. - Calculate equilibrium conditions between two phases of a pure component (i.e. water-vapour equilibrium). - Calculate equilibrium vapor pressures over multicomponent liquids. - Calculate Chemical equilibrium of reaction mixtures. Attitude at the end of the course, the student is familiar with - The general principles of thermodynamics - applying these principles to processes in the atmosphere, ocean and the solid earth.	
Education Method	The course will consist of lectures and assignments and exercises, using a text book and computer-aided assignments. Most lectures will cover one particular topic, although larger topics will require 2 lectures to discuss. From students it is expected that they have studied the lecture topic from the book in advance of the lecture, and complete the assigned exercises for each week before start of the following week's lecture. Extra training opportunities are offered during weekly exercise trainings.	
Assessment	Weekly Assignments. 1 Final exam	
Expected prior Knowledge	Principles of Chemistry, Mechanics. Calculus 1&2	
Academic Skills	Differential Equations, Problem Solving Group Work,	
Literature & Study Materials	There is no perfect book dealing with thermodynamics for geoscience. We will roughly follow the chapters of the book Physical Chemistry of Thomas Engel and Philip Reid (2nd edition) . Please make sure that you have a electronic or a hardcopy of this book to your disposal. In addition, material will be used from lecture notes and from the book: Thermodynamics: From Concepts to Applications of Arthur Shavit & Chaim Gutfinger 2nd edition. This material will be made available for you in Brightspace.	
Judgement	Weekly Assignments. 1 Final exam For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Pen or pencil, paper, basic (non-graphing) calculator. The necessary tables and formulas will be provided at the exam. The use of own books with tables is not allowed.	
Collegerama	No	

AESB2220-20 Toets 1	Final exam / Lab /Midterm	5
Responsible Instructor	Dr. A.P. Siebesma	
Contact Hours / Week x/x/x/x	4 (lect.) + 1 every 2 weeks	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See AESB2220-20	
Study Goals	See AESB2220-20	
Education Method	See AESB2220-20	
Assessment	See AESB2220-20	
Expected prior Knowledge	See AESB2220-20	
Academic Skills	See AESB2220-20	
Literature & Study Materials	See AESB2220-20	
Judgement	See AESB2220-20	
Permitted Materials during Exam	See AESB2220-20	
Collegerama	No	

AESB2220-20 Toets 2	Assignments	0
Responsible Instructor	Dr. A.P. Siebesma	
Contact Hours / Week x/x/x/x	4 (lect.) + 1 every 2 weeks	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	See AESB2220-20	
Study Goals	See AESB2220-20	
Education Method	See AESB2220-20	
Assessment	See AESB2220-20	
Expected prior Knowledge	See AESB2220-20	
Academic Skills	See AESB2220-20	
Literature & Study Materials	See AESB2220-20	
Judgement	See AESB2220-20	
Permitted Materials during Exam	See AESB2220-20	
Collegerama	No	

AESB2230	Sedimentology & Reservoir Geology	5
Responsible Instructor	Dr. H.A. Abels	
Contact Hours / Week x/x/x/x	2hrs lecture p/week 4hrs practical p/week	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This course is an introduction to applied sedimentology, sedimentary environments, sedimentary processes and facies, sediment preservation possibilities and resulting sedimentary architecture in the subsurface. It covers sediment-petrographic analysis of sandstones and carbonate rocks, relationship between sediment-petrographic parameters and petrophysical properties, and the depositional environment.	
Course Contents Continuation	The course Sedimentology & Reservoir Geology is an introduction to these topics: The various sedimentary environments are analyzed and related sedimentary processes discussed. The resulting sedimentary facies characteristics are discussed, with special emphasis on: - The preservation possibilities and controls of sedimentary facies; - The internal structures and heterogeneities of sedimentary bodies; - The external shape of sedimentary bodies; - The architecture of sedimentary bodies in the basin fill and its controlling parameters; - Subsurface (well log) expression of the preserved sediments The practical introduces the students in sediment-petrographic analysis, using cuttings and thin sections of cores and rock samples. The practical is focused on the description of sands, sandstones and carbonate rocks. The emphasis in the description is on the relation between the different sediment-petrographic parameters (like grain size, sorting, texture, diagenesis) and petrophysical properties that relate to porosity and permeability.	
Study Goals	After following this course, students are able to: - reconstruct sedimentary processes and environments through the analysis of sedimentary facies descriptions - describe the lateral and vertical variability as well as internal heterogeneities of sediment bodies related to different sedimentary environments and internal and external controls - discuss the reservoir properties of these 3D sedimentary bodies in the subsurface and predict 3D sedimentary character in the subsurface based on limited data After the practical, students are able to: - identify sedimentary facies of carbonates and sandstones using microscopic description of cuttings and thin sections - relate the facies descriptions with downhole geophysical log properties - identify and discuss the reservoir rock properties of sedimentary units using sedimentary and petrophysical data	
Education Method	The course consists of lectures and practicals. The course consists of eight classes of two hours. The course starts with a two-hour introduction in the discipline of Sedimentology and its relation to other courses in the curriculum. Principles and definitions are explained, and the methodology of sedimentary facies analysis in outcrop and the sub-surface is introduced. The remaining classes deal with the facies characteristics of different sedimentary environments. During the lectures, case studies related to reservoir geology will be discussed and the predictive methodology of sediments in the subsurface discussed. The microscopy practical occurs in sessions of three hours. It starts with a general recap of theory and guideline of the thin section microscopy of rocks. After every practical, the students hand-in (individually or in groups of two) an assignment. 2hrs lecture p/week 4hrs practical p/week If Covid-19 measures are valid, the course consists of online lectures, home work and, when allowed, physical practicals. The course consists of eight physical or online classes of two hours. The course starts with a two-hour introduction in the discipline of Sedimentology and its relation to other courses in the curriculum. Principles and definitions are explained, and the methodology of sedimentary facies analysis in outcrop and the sub-surface is introduced. The remaining classes deal with the facies characteristics of different sedimentary environments. During the lectures, case studies related to reservoir geology will be discussed and the predictive methodology of sediments in the subsurface discussed. The microscopy practical occurs in 8 sessions of two hours. It starts with a general recap of theory and guideline of the thin section microscopy of rocks. After every practical, the students hand-in (individually or in groups of two) an assignment that requires some homework on the materials. 2hrs lecture p/week physical / online 2hrs practical p/week on campus 2hr practical p/week homework	
Course Relations	You should have passed Geology 1, 2 & 3 (AESB1130, AESB1230, AESB1430) and the course is a pre-requisite for the Geological Fieldwork (AESB2430 & AESB2431)	
Literature and Study Materials	Textbook plus additional literature posted on Brightspace plus the Power-Point files of the course(also available on Brightspace). The textbook has to be acquired (via MV) prior to the start of the course: Sedimentology and Stratigraphy (2nd Edition) by Gary Nichols, Wiley-Blackwell,ISBN: 1405193794	
Prerequisites	Geology 1, 2 & 3 (AESB1130, AESB1230, AESB1430)	
Assessment	The course will be assessed by 2 separate parts that both need to be graded a 5 as minimum to pass; Theory and Practical Theory (50% of Final grade; minimum 5): - Theoretical examination on lecture content (40%) - Written report or presentation on own sedimentary research item (10%) Practical (50% of Final grade; minimum 5): - Practical examination (30%) - Practical Exercises (20%) The contents of the introductory lectures, the practical manual and the notes taken during the practical, constitute the subject matter of the examination. The practical examination will take place soon after the course.	

	--- UPDATE dd 6-11-2020 due to the COVID-19 measures --- The course will be assessed by 2 separate parts that both need to be graded a 5 as minimum to pass; Theory and Practical Theory (50% of Final grade; minimum 5): - Theoretical examination on lecture content (40%) - Written report or presentation on own sedimentary research item (10%) Practical (50% of Final grade; minimum 5): - Practical examination (30%) - Practical Exercises (20%) The contents of the introductory lectures, the practical manual and the notes taken during the practical, constitute the subject matter of the examination. Both examinations will be online in Brightspace Assignments or Quizzes.
Contact	Dr. H.A. Abels h.a.abels@tudelft.nl +31 (0)15 27 85722
Expected prior Knowledge	Geology 1, 2 & 3 (AESB1130, AESB1230, AESB1430)
Academic Skills	Insight in 3-D problems, develop a small research project
Literature & Study Materials	Textbook plus additional literature posted on Brightspace plus the Power-Point files of the course
Judgement	Written exams, report, presentation, assignments For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	none
Collegerama	No

AESB2230 Toets 1	***Written exam***	***2.5***
Responsible Instructor	Dr. H.A. Abels	
Contact Hours / Week x/x/x/x	see description AESB2230	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	see description AESB2230	
Study Goals	see description AESB2230	
Education Method	see description AESB2230	
Assessment	see description AESB2230	
Expected prior Knowledge	See: course description AESB2230	
Academic Skills	See: course description AESB2230	
Literature & Study Materials	See: course description AESB2230	
Judgement	See: course description AESB2230	
Permitted Materials during Exam	See: course description AESB2230	
Collegerama	No	

AESB2230 Toets 2	*Practical exam*	*2.5*
Responsible Instructor	Dr. H.A. Abels	
Contact Hours / Week x/x/x/x	see description AESB2230	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	see description AESB2230	
Study Goals	see description AESB2230	
Education Method	see description AESB2230	
Assessment	see description AESB2230	
Expected prior Knowledge	see description AESB2230	
Academic Skills	see description AESB2230	
Literature & Study Materials	see description AESB2230	
Judgement	see description AESB2230	
Permitted Materials during Exam	see description AESB2230	
Collegerama	No	

AESB2320	Physical Transport Phenomena	5
Responsible Instructor	Dr. A.P. Siebesma	
Responsible Instructor	Drs. J.C. Blom	
Contact Hours / Week x/x/x/x	4-6hrs lecture p/week (46hrs total)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Math 1, 2, 3 & 4(one and multiple variables, differential equations) (AESB1110, AESB1210, AESB1310, AES2110) Mechanics 1 (AESB1320), Mechanics 2, Electricity & Magnetism (AESB1420), Chemical Thermodynamics (AESB2220)	
Course Contents	The first part of the course will be concerned with fluid mechanics, culminating with an exam on part 1; and the second part heat transfer and mass transfer, with a separate exam on Part 2. This course is designed to cover essential aspects of fluid flow, heat transfer and mass transfer, with special attention to topics important to applied earth sciences. First, aspects of fluid flow will be treated, among them laminar flow, turbulence, and dimensionless correlations for drag in laminar and turbulent flow. Subsequently the attention is focused on heat transfer. Covered subjects include shell balances for simple heat-transfer problems, conduction through multiple layers, unsteady conduction, convective heat transfer, approaches to simplifying transport problems, and radiation. Finally mass transfer is treated briefly, with attention on unsteady diffusion and convection mass transfer.	
Study Goals	The aim of the course is to acquire knowledge about processes of transport of mass, energy and momentum. In this course attention will be paid to the various applications of transport phenomena in engineering and everyday life, with particular attention to processes important to applied earth sciences.	
Study Goals continuation	At the end of this course, the student is able to do the following when presented with a problem: - Designate the main transport mechanisms - Establish an appropriate balance for this problem - To model the various forms of transport - On the basis of the above, to set up a relevant description / modeling of the problem - If requested, give numeric answers to detailed questions.	
Education Method	Teaching method will be primarily lecture; as time allows, some lecture hours will be given over to working and going over homework problems. Students should spend the self-study time first reviewing the content of lectures, and then working practice problems using the methods and approaches shown in the lectures. --- UPDATE dd 2-2-2021 due to the COVID-19 measures --- The lectures are available on Collegerama. Students should watch the lectures on their own and take notes. A schedule of which lectures should be watched each week during the course will be distributed to the class. Once per week the class will meet to review of the material covered, ask questions, do practice problems and prepare for the exams. These meetings will be on campus if allowed by campus regulations.	
Course Relations	Connections will be made between fluid mechanics in physical transport phenomena and in Mechanics and Transport by Flow in Porous Media (AESB3340). Taught simultaneously with Soil Mechanics; some connection will be made between normal and shear stress in fluid mechanics and in soil and solid mechanics.	
Literature and Study Materials	Text: R. B. Bird, W. E. Stewart, E. N. Lightfoot, & D. J. Klingenberg, Introductory Transport Phenomena, John Wiley & Sons, New York, 2015, with essential supplemental material from First Edition of Bird, Stewart and Lightfoot, Transport Phenomena, Wiley, 1960. Text MUST be purchased from Mijnbouwkundige Vereeniging to receive supplemental material. Plus handouts distributed in class and made available on the Brightspace web site.	
Assessment	This course is taught in two parts. At the end of each part there is a written exam on that part. Exams will be given in weeks 5 and 10. Re-exams will be given for each of the separate exams. The higher of the two scores on a given part-exam will count toward the grade. Judgement: A student must receive a score of at least 5 on each part exam to pass. The final grade is then the average of the two part-exam grades (50/50). --- UPDATE dd 2-2-2021 due to the COVID-19 measures --- This course is taught in two parts. At the end of each part there is a written exam on that part. Exams will be given in weeks 5 and 10. Exams will be administered remotely if required by campus regulations. Re-exams will be given for each of the separate exams. The higher of the two scores on a given part-exam will count toward the grade.	
Permitted Materials during Tests	In the examinations, the text, class notes and a limited number of extra pages of notes (but not the extra worked exercises on the Brightspace site or old examinations) will be allowed, as will be announced before the exam. Students may bring a calculator to the exam but no computers or communications devices (e.g., phone or tablet computer).	
Contact	W.R. Rossen w.r.rossen@tudelft.nl tel: +31 (0)15 27 86038	
Expected prior Knowledge	Math 1, 2, 3 & 4(one and multiple variables, differential equations) (AESB1110, AESB1210, AESB1310, AES2110) Mechanics 1 (AESB1320), Mechanics 2, Electricity & Magnetism (AESB1420), Chemical Thermodynamics (AESB2220)	
Academic Skills	See expected prior knowledge.	
Literature & Study Materials	Text: R. B. Bird, W. E. Stewart, E. N. Lightfoot, & D. J. Klingenberg, Introductory Transport Phenomena, John Wiley & Sons, New York, 2015, with essential supplemental material from First Edition of Bird, Stewart and Lightfoot, Transport Phenomena, Wiley, 1960. Text MUST be purchased from Mijnbouwkundige Vereeniging to receive supplemental material. Plus handouts distributed in class and made available on the Brightspace web site.	
Judgement	A student must receive a score of at least 5 on each part exam to pass. The final grade is then the average of the two part-exam grades (50/50). For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	In the examinations, the text, class notes and a limited number of extra pages of notes (but not the extra worked exercises on the Brightspace site or old examinations) will be allowed, as will be announced before the exam. Students may bring a calculator to the exam but no computers or communications devices (e.g., phone or tablet computer).	
Collegerama	Yes	

AESB2320 Toets 1	*Exam 1*	*2.5*
Contact Hours / Week x/x/x/x	see description AESB2320	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	see description AESB2320	
Study Goals	see description AESB2320	
Education Method	see description AESB2320	
Assessment	see description AESB2320	
Expected prior Knowledge	see description AESB2320	
Academic Skills	see description AESB2320	
Literature & Study Materials	see description AESB2320	
Judgement	see description AESB2320	
Permitted Materials during Exam	see description AESB2320	
Collegerama	Yes	

AESB2320 Toets 2	*Exam 2*	*2.5*
Contact Hours / Week x/x/x/x	see description AESB2320	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	see description AESB2320	
Study Goals	see description AESB2320	
Education Method	see description AESB2320	
Assessment	see description AESB2320	
Expected prior Knowledge	see description AESB2320	
Academic Skills	see description AESB2320	
Literature & Study Materials	see description AESB2320	
Judgement	see description AESB2320	
Permitted Materials during Exam	see description AESB2320	
Collegerama	Yes	

AESB2330	Soil Mechanics	5
Responsible Instructor	Prof.dr. M.A. Hicks	
Instructor	Dr. P.J. Vardon	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The course is split into 2 parts: the first part is largely equivalent to the old Soil Mechanics 1 course; the second part is largely equivalent to the old Soil Mechanics 2 course.	
Summary	The course explains the basic concepts of theoretical and applied soil mechanics. Theoretical soil mechanics includes: Soil Characteristics; Groundwater; Geomechanics; Shear Strength of Soils. Applied soil mechanics includes: Retaining Structures; Foundations; Slope Stability.	
Course Contents	Soil Characteristics: Classification and mineralogy of soils; Grains, grain size distribution and porosity; Introduction to in situ testing. Groundwater: Pore pressure and effective stress; Darcys law, permeability and groundwater flow. Geomechanics: Stresses and initial stress state; Strains, stress-strain relationships and tangent modulus; Elastic solutions; Consolidation, drained and undrained behaviour. Shear Strength of Soils: Drained and undrained soil behaviour; total and effective shear strength parameters; Mohr circles; Mohr-Coulomb failure criterion; stress paths; shear test; triaxial test; site investigation; soil sampling. Retaining Structures: Lateral earth pressure at rest; passive and active pressures; Rankine; Coulomb; sheet pile walls; Blum. Foundations: Geotechnical bearing capacity; shallow and deep foundations; Prandtl; Brinch Hansen; pile foundations. Slope Stability: Limit equilibrium methods; vertical slope; infinite slope; method of slices; Fellenius; Bishop.	
Study Goals	The main goal of this course is to: Have an understanding of, and be able to apply, the following: Soil classification; In situ testing; Groundwater flow; Stresses in the ground; Strains and stiffness; Consolidation theory; Shear strength of soils; Laboratory and in situ testing; Earth retaining structures; Foundations; Slope stability. By the end of the course the student should be able to: Explain the various forms of soil behaviour, including the influence of soil properties, degree of saturation and loading rate. Evaluate laboratory data to determine material parameter values. Plan an engineering strategy for analysing geotechnical problems, involving settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability. Analyse appropriately and evaluate geotechnical problems, utilising presented theory, on the following topics: elastic behaviour and settlements, consolidation, groundwater flow, basic foundations, retaining structures and slope stability. Study load: 36 lecture hours, 12 tutorial hours, 72 self-study hours, 17 exam preparation hours, 3 exam hours.	
Education Method	Lectures, tutorials and self-study.	
Course Relations		
Assessment	The learning goals of this course will be tested with: Written examination (in English) at the end of the course. Mid-term test must be passed to gain entry to the exam. Judgement: Mid-term test must be passed to gain entry to the exam. Final mark = examination grade only.	
Expected prior Knowledge	A knowledge of mechanics.	
Academic Skills	Analytical skills.	
Literature & Study Materials	Course book (optional): Soil Mechanics, A. Verruijt, Delft University of Technology, 2012. Go to: http://geo.verruijt.net/ and download the book (in English) stored as SoilMechBook2012.pdf Accompanying lectures notes (in the form of ppt slides). Worked examples.	
Judgement	Mid-term test must be passed to gain entry to the exam. Final mark = examination grade only. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Standard calculator. Formula sheet will be provided.	
Collegerama	Yes	

AESB2342	Rock Mechanics	5
Responsible Instructor	Dr.ir. A.A.M. Dieudonné	
Contact Hours / Week x/x/x/x	48 hours over 9 weeks	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The main goal of this course is to understand and be able to apply the principles of rock mechanics to a range of rock engineering applications, including the design of underground and surface excavations. The course is divided in 7 modules: 1. Concepts of stress and strain 2. Behaviour of intact rock 3. Behaviour of discontinuities 4. Rock mass and rock mass classifications 5. Underground excavations 6. In situ stress 7. Surface excavations	
Study Goals	On successful completion of this course the student is able to: [LO1] Interpret laboratory test results on rock samples and rock discontinuities, select an appropriate constitutive model and determine its parameter values. [LO2] Evaluate the stability of underground and surface excavations with simple geometries, by means of simple models. [LO3] Characterise a rock mass and explain the role of discontinuities on the design of excavations, by means of qualitative and quantitative arguments. [LO4] Extract the relevant information from available data to solve a rock engineering problem. [LO5] Integrate aspects of safety in rock engineering and evaluate options for mitigating rock excavation instabilities. In the above-mentioned learning objectives, excavations refer to both underground (shafts, boreholes, tunnels, mines...) and surface (slopes, mines...) excavations.	
Education Method	The course follows a blended learning approach that combines self-study activities and face-to-face lectures (both online and on campus). Case studies are used throughout the course to apply or demonstrate theoretical concepts.	
Assessment	- Group assignments (45% of the final grade) - Open-book written exam on the whole course content (55% of the final mark) - AES Information Literacy 2 module (Pass/fail)	
Expected prior Knowledge	- Mechanics (AESB1320-17) - Mathematics 1 to 5 - Instrumentation & Signals with Python (AESB2122)	
Academic Skills	See Study Goals.	
Literature & Study Materials	All course materials (lecture notes, articles, videos, etc.) are given on the Brightspace site.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/	
Permitted Materials during Exam	The exam is an open-book exam, the student is allowed to bring all books, lecture notes and worked examples. No electronic media with wireless or network capabilities are allowed as the exam is an individual assessment and no collaboration is allowed.	
Collegerama	No	

AESB2430	Geological Fieldwork: Data Acquisition	5
Responsible Instructor	Drs. J.C. Blom	
Instructor	Dr. R.C. Lindenbergh	
Responsible for assignments	Dr. H.A. Abels	
Responsible for assignments	Dr. J.E.A. Storms	
Contact Hours / Week x/x/x/x	In total 140 hours of which 3 weeks full-time fieldwork in the SE of France and 20 hours preparation in Delft	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Course Contents	Classroom instruction: introduction to the fieldwork, lectures on geological mapping techniques and cross-section construction Fieldwork: basic field observations and interpretations, mapping and sampling techniques, use of remote sensing techniques, aerial photographs and topographic base maps, compiling a data report with accompanying maps and cross-sections in accordance with international standards.	
Study Goals	To perform a geologic field project and to collect and present geological data	
Study Goals continuation	At the end of this course students will be able to: - obtain an understanding of geologic phenomena - perform a geologic mapping project - complete a project in a team - present the acquired data in a form suitable for continued analysis - obtain understanding of his/her personal learning methods and those of others - obtain an independent attitude	
Education Method	Classroom instruction, 3 weeks fieldwork in France, self-study Classroom instruction with associated exercises, fieldwork. The fieldwork will be conducted in a team of their own choice, consisting of three students, and will cover an area of roughly 50 km2 in the Subalpine Ranges in the Drome department in France. During the fieldwork the students will be camping, and responsible for their own transport and meals.	
Course Relations	Geostatistics & Remote Sensing (AESB2440) has a practical part that takes place during the field work, see prerequisites	
Prerequisites	Geology 1 (AESB1130), Geology 2 (AESB1230), Geology 3 (AESB1430), Sedimentology & Reservoir Geology (AESB2230). AESB2440 needs to be followed in the same period or must have been completed earlier.	
Assessment	Pre-fieldwork: assignments, to be handed in before the fieldwork During fieldwork: - a progress meeting after about 10 days of field work, - grades on participation by all supervisors - two written and hand-drawn data reports, , one on the geology, one on the remote sensing excersises, to be handed in on the last day of the fieldwork. Judgement: Grades will be given to the following parts of the course: - the pre-fieldwork excercises - the intermediate progress meeting during the field work - the participation during the field work, as judged by the supervisors - the two data reports (geology & remote sensing) handed in at the end of the field work All five parts will count equally for the final grade. There is no resit: if students fail, students have to redo the fieldwork at their own expense	
Contact	Drs. J.C. Blom j.c.blom@tudelft.nl tel: +31 (0)15 27 83628 Dr. R. Lindenbergh r.lindenbergh@tudelft.nl tel: +31 (0)15 27 87649	
Expected prior Knowledge	Geology classes from the first year: AESB1130, 1230 & 1430 Geology classes from the second year: AESB2230	
Academic Skills	During the field work, the idea is that the students apply the knowledge the acquired during their theoretical classes on geology and remote sensing to a practical problem (their fieldwork area) and use this knowledge to analyse the geologic structure of their areas. They are then asked to synthesise their observations with the knowledge gained from external sources like literature and sattelite images, and to evaluate these results.	
Literature & Study Materials	All students need a hand lens, a geologic compass. a fieldbook and a geologic hammer. In addition, sturdy boots/shoes and a reflective vest are mandatory A4 and A3 transparant tracing paper will be extremely useful. Necessary literature will be made available via Brightspace.	
Judgement	Grades will be given to the following parts of the course: - the pre-fieldwork excercises - the intermediate progress meeting during the field work - the participation during the field work, as judged by the supervisors - the two data reports (geology & remote sensing) handed in at the end of the field work All five parts will count equally for the final grade. There is no resit: if students fail, students have to redo the fieldwork at their own expense. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during	not applicable	

Exam

AESB2431	Geological Fieldwork: Data Integration	5
Responsible Instructor	Drs. J.C. Blom	
Instructor	Dr. R.C. Lindenbergh	
Contact Hours / Week x/x/x/x	20hrs contact 120hrs self-study	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Course Contents	In this course, students will work in small groups to produce a scientific/technical report consisting of two parts. In the first part of the report, data gathered during the geological fieldwork will be placed in broader scientific framework provided by background information from the scientific literature. The second part of the report is devoted to integration of fieldwork data with RS data for the purpose of empirical classification, in order to evaluate the extent to which automated mapping of the fieldwork area is possible. This part also includes prediction of the geological structure of a fieldwork area mapped by another group of students. Each group receives written feedback from the group whose fieldwork area has been evaluated based on RS data alone, and is expected to reflect on this peer assessment in written form. This course effectively forms one course together with AESB2430	
Study Goals	To process spatial data (remotely sensed data as well as fieldwork data collected in course AESB2431), to research background literature on the geological evolution of the area under investigation, and to describe the workflow and discuss its findings in a scientific report.	
Study Goals continuation	At the end of this course students are able to: - Integrate (their own) geological field observations with Remote Sensing (RS) data - Process spatial data to make predictions in similar, but unknown terrains. - Carry out literature research for the purpose of placing their own results in a broader scientific framework. - Write a scientific report according to generally accepted guidelines. - Asses the work of other groups and give feedback to peers, as well as receive feedback from peers.	
Education Method	Students will work in small groups (in principle, the same as in AESB2430). Classroom instruction; feedback from instructors & peer reviews to be received on drafts of specific chapters/topics (e.g. synthesis of geological background literature).	
Course Relations	Geostatistics & Remote Sensing (AESB2440), Field Exploration Project (AESB3440), see prerequisites	
Literature and Study Materials	a list of suggested articles is supplied to start the bibliographic research	
Prerequisites	Sedimentology & Reservoir Geology (AESB2230), Geological Fieldwork Data Aquisition (AESB2430), Statistical Spatial Data Aanlysis (AESB2440)	
Assessment	Written report, comprising analysis of the fieldwork data gathered in AESB2431 in relation to remotely sensed data and information drawn from the scientific literature. Also included is a prediction of the geology in a different field area based on remote sensing data calibrated on the students own fieldwork area. Groups will receive marks based on quality of early drafts (25%), peer assessment and feedback (25%), and final report (50%).	
Contact	Drs. J.C. Blom j.c.blom@tudelft.nl tel: +31 (0)15 27 83628 Dr. R. Lindenbergh r.lindenbergh@tudelft.nl tel: +31 (0)15 27 87649	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

AESB2440	Geostatistics & Remote Sensing	5
Responsible Instructor	Dr. R.C. Lindenbergh	
Contact Hours / Week x/x/x/x	2x10hrs lecture 2x10hrs exercise classes	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Lecture 1: Course overview; Digital mapping: Geoids, local and global coordinate systems, coordinate transformations; Map projections; Remote Sensing data: Spectral bands, Sentinel-2 data, Photogrammetry, Global Digital Elevation Models, Freely available Remote Sensing data Lecture 2: Recap on Probability, Events, Discrete and continuous sample spaces, Lecture 3: Sensors: GNSS; Errors in data; Histogram; Inverse distance interpolation Lecture 4: Deterministic vs. stochastic interpolation, distance, spatial continuity, Convex hull, Voronoi diagrams, Delaunay triangulation, Triangulated Irregular Networks, Lecture 5: Recap on Conditional probability, Total probability rule, Bayes rule, Independence, Random variable, distribution function, probability mass function, Distributions: binomial, uniform, normal, standard normal, transformation to standard normal. Lecture 6: Least squares: mean, variance, covariance, correlation, Random function, minimizing the least squares error, least squares geometry Lecture 7: Stochastic interpolation. Kriging the mean, Simple Kriging equations, experimental covariance function, Positive definiteness, variogram, Sill, Range, Nugget. Lecture 8: Terrain classification. Band data, false colour images, feature space, Supervised classification, training data, decision tree, linear separation, nearest centroid classification, Spectral angular mapping, Maximum likelihood classification. Unsupervised classification, K-means clustering, Confusion matrix Lecture 9: Digital Terrain Analysis. plane fitting, variance, roughness, surface normals, slope, aspect, orientation, gradient, drainage network, D8, flow accumulation, watersheds Lecture 10: Quality assessment. Robust methods, Outliers, median, MAD, RANSAC, Monte Carlo simulations.	
Study Goals	At the end of this course you will be able to critically derive information from uncertain geospatial data.	
Study Goals continuation	The student will acquire both methodological and practical skills to analyse uncertain spatial data. After this course the student will be able to: - apply different statistical techniques to assess spatial continuity and inherent uncertainty of spatial measurements and derived parameters. - explain the differences between a number of acquisition techniques, such as notably GNSS, LIDAR, photogrammetry and spectral remote sensing. - analyze remote sensing and surveying data using a GIS system and to distinguish between analysis methods like interpolation and classification in terms of computational efficiency, robustness, ease of use and quality of the results.	
Education Method	Lectures, exercise classes, assignments, one afternoon of data gathering. During the exercise classes, students will work on exercises and assignments. Students will be asked and assisted to install the QGIS software on their own laptop. Students will analyse remote sensing data around Delft and sampling the Geological fieldwork area (AESB2430/AESB2431). Students will gather in situ elevation and thematic data in around Delft using their Smartphone as sensor	
Course Relations	This course is directly related to the 2nd years Geological Fieldwork, that runs in the same quarter. Considered data will be from the fieldwork area, while methods introduced in the course will be applied by the students during their fieldwork analysis.	
Prerequisites	Probability & Statistics, AESB1213 Signals & Systems with Python (AESB2122), Math 1 & Linear Algebra	
Assessment	50 % Assignments 50 % Final written exam The written exam will take place in the exam period at the end of the term; Assignment deadlines are planned shortly after corresponding lectures. The final schedule will be adapted to the fieldwork planning	
Contact	Dr. R.C. Lindenbergh, r.c.lindenbergh@tudelft.nl tel: +31 (0)15 27 87649	
Expected prior Knowledge	* Theory and skills from Calculus, Linear Algebra & Probability and Statistics * Basic programming skills in for example Python	
Academic Skills	Academic writing (assignments); Cooperation (assignemnts are in groups of two); Planning (Course goes very fast); Problem analysis; (Assignments have an open nature) Connecting theory to practice; (Theory is directly applied in practical assignments)	
Literature & Study Materials	Background material - Modern Introduction to Probability and Statistics, (2005), F. M. Dekking et al., Springer (available as TU Delft Springer Link) - Photogrammetric Computer Vision, Ch.2 and Ch.4, W. Foerstner & B. Wrobel, (2016), Springer (available as TU Delft Springer Link) - Basic Steps in Geostatistics: The Variogram and Kriging,	

	(2015), Margaret A. Oliver, Richard Webster, SpringerBriefs in Agriculture, Springer (available as TU Delft Springer Link) - Primer on Mathematical Geodesy, C. Tiberius, Course notes - Primer on Signals and Measurements, C. Tiberius, Course notes - Primer on Reference systems for surveying and mapping, H. van der Marel, Course Notes
Judgement	50% assignments, 50% final written exam; For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Students are allowed to use a simple pocket calculator without memory
Collegerama	No

AESB2440 Toets 1	***Exam***	***2.5***
Responsible Instructor	Dr. R.C. Lindenbergh	
Contact Hours / Week x/x/x/x	see description AESB2440	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	see description AESB2440	
Study Goals	see description AESB2440	
Education Method	see description AESB2440	
Assessment	see description AESB2440	
Expected prior Knowledge	see description AESB2440	
Academic Skills	see description AESB2440	
Literature & Study Materials	see description AESB2440	
Judgement	see description AESB2440	
Permitted Materials during Exam	see description AESB2440	
Collegerama	No	

AESB2440 Toets 2	***Exercises***	***2.5***
Responsible Instructor	Dr. R.C. Lindenbergh	
Contact Hours / Week x/x/x/x	see description AESB2440	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	see description AESB2440	
Study Goals	see description AESB2440	
Education Method	see description AESB2440	
Assessment	see description AESB2440	
Expected prior Knowledge	see description AESB2440	
Academic Skills	see description AESB2440	
Literature & Study Materials	see description AESB2440	
Judgement	see description AESB2440	
Permitted Materials during Exam	see description AESB2440	
Collegerama	No	

Year	**2022/2023**
Organization	**Civil Engineering and Geosciences**
Education	**Bachelor Applied Earth Sciences**

AES BSc 3e jaar 2022

Year	**2022/2023**
Organization	**Civil Engineering and Geosciences**
Education	**Bachelor Applied Earth Sciences**

AES BSc 3e jaar Minor Programma

Year	**2022/2023**
Organization	**Civil Engineering and Geosciences**
Education	**Bachelor Applied Earth Sciences**

AES BSc 3e jaar Major Programma

AESB3340	Mechanics and Transport by Flow in Porous Media	5
Responsible Instructor	S. Muraro	
Course Coordinator	S. Muraro	
Contact Hours / Week x/x/x/x	0/0/4/0 Asynchronous feedback on discussion board 1x2 hour lecture on computer assignment Asynchronous feedback on computer assignment (Brightspace) Fixed weekly office hours for feedback per group	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course, students acquire a fundamental basis for one- and two-phase flow in porous media. This course is an essential basis for all tracks within the MSc Applied Earth Sciences as well as the MSc track Geo-engineering in the MSc Civil Engineering. This course is highly recommended for students interested in Hydraulic Engineering as well as Hydro-geology and Water Management. The course will address the following topics: - basic principles for groundwater flow under saturated conditions (1D, 2D and 3D, both stationary as well as non-stationary flow); - (potential) energy and its relationship with pressure and head; - capillary pressure and unsaturated flow (2 phase flow in a porous medium filled with water and air); - concepts of water and air content, saturation, hydraulic conductivity, relative permeability and residual saturation; - water retention curves (van Genuchten, Brooks-Corey, and Gardner equations); - the impact of flow and transport on soil mechanics, highlighting the connection with the parallel course of soil mechanics.	
Study Goals	At the end of this course you will be able to: - explain the origins of groundwater flow; - derive the governing equations of the groundwater flow with appropriate initial and boundary conditions for saturated soils and unsaturated soils; - solve laboratory and field steady-state and transient problems for saturated (1D, 2D) and unsaturated (1D) flow by programming a Python/Matlab script; - discuss the quality and the limitations of the model results.	
Education Method	The course is structured along weekly lectures and e-tivities (so-called online activities). The aim of this structure is to provide students with valuable feedback of their progress. The content is structured with a study guide which is based on topics discussed in a number of standard handbooks. - Self-study The key to passing this course is self-study. The heart of the course is a study guide. In this guide all topics are specified in based on assignments which the students are required to make. The study guide points to the relevant chapters and paragraphs in a number of handbooks which the students need to study in order to be able to solve the assignments. There are several very good reference books available that cover the content of this course. - Lectures The study guide describes the content of each lecture and the students are expected to prepare for each lecture by making assignments and reading the required background material. The course is organised along so-called e-tivities, activities integrated in Brightspace. These e-tivities are a series of test assignments with which the students can evaluate their progress themselves. Students are expected to upload questions to Brightspace about the assigned work or any subjects they have difficulties with. During the lectures these questions will be addressed. The lectures will be interactive and led by student questions and student feedback. - Asynchronous feedback and dialog Using the Discussion Board on Brightspace environment, students are expected to collaborate in order to identify questions and to help each other complete the assignments. These tools are also used to give and get feedback on intermediate results (i.e. the assignments) but also on lectures etc. - Portfolio of individual computer assignments Students in the Applied Earth Sciences program are expected to make a portfolio of computer assignments in which they show that they are able to use the theoretical approaches from the literature to solve the problems given in the assignments. The aim of the portfolio is to develop understanding and show that the student is able to apply the theory under different scenarios. The portfolio provides a summary of the self-study activities performed by the students during the course. The progress of the portfolio will be monitored using Brightspace. The final version of the portfolio must be submitted before the exam. --- ONLINE TEACHING IN CASE OF COVID-19 measures --- In the case of COVID-19 restrictions to on-campus lectures, the course will be given online as in 2021.	
Course Relations	This course is closely related and has some overlap with Soil Mechanics (CTB2310) and Physical Transport Phenomena (AESB2320). The course is fundamental to the Applied Earth Sciences field and integrates different aspects with respect to the flow of water through porous media.	
Literature and Study Materials	Study guide, lecture notes, material on Brightspace, links to several good handbooks.	
Assessment	Final exam in week 10. The exam consists of a series of questions that are similar to the assignments prepared for and discussed during the lectures. The online activities and portfolio given during the course are compulsory activities and enter in the final summative assessment (after the final exam is passed). The final grade for AES is calculated as: 80% (70% exam + 30% Online Activities) + 20% Portfolio --- ONLINE FINAL EXAM IN CASE OF COVID-19 measures --- In the case of COVID-19 restrictions to on-campus exam, the exam will consist of online assignments in Brightspace. Each set of questions will open at a specific time and will close at a specific time. The exam is open book and consists of open-ended questions. Multiple versions of the exam questions will be used. Answers to the questions need to be uploaded to Brightspace as word or pdf files.	

	A random selection of students will be contacted for a short oral check related to their exams.
Permitted Materials during Tests	No intermediate tests will be given.
Contact	Dr. Stefano Muraro Email S.Muraro@tudelft.nl Phone +31 15 27 83327
Expected prior Knowledge	All Math, Mechanics 1, Soil Mechanics
Academic Skills	Knowledge, students: - know the concepts and assumptions which form the basis of Darcy's Law; - are able to apply Darcy's law in a range of different applications; - understand how to simplify the 3D flow equation to 2D or 1D; - are able to explain when the assumptions used in the reduction of dimensionality are applicable; - understand the concepts of capillary pressure and unsaturated flow; - understand and be able to use the van Genuchten, Brooks-Corey, and Gardner equations; Skills, students: - are able to derive quantitative solutions to a range of engineering problems using analytical solutions; - have developed skills to share information and knowledge in an online environment; - have developed skills to help others in an online environment; - have developed the skill to use Python/Matlab to assess simple numerical and analytical solutions to flow and transport problems; - will have the skill to integrate transport of solutes with the groundwater flow problems in order to analyse groundwater contamination problems; Attitude, students: - will have developed an appreciation of the intrinsic value of the sub-surface for society, ecosystem etc. - will have developed an open attitude for collaboration in an online environment.
Literature & Study Materials	All course materials are given on the Brightspace site. The course is based on the following books: Groundwater Science, 2013, Charles R. Fitts, second edition, Academic Press, Elsevier. Unsaturated Soil Mechanics, 2004, Lu, Ning; Likos, William J., John Wiley & Sons: Chapters 2, 3, 4, 8, 9. You are strongly advised to buy the book by Fitts. For Lu and Likos, we only use a small number of chapters. Printed copies of downloads will suffice.
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	ON-CAMPUS FINAL EXAM The exam is an open book exam, the student is allowed to bring all books, lecture notes, and worked examples. The student is allowed to use an electronic calculator. No electronic media with wireless or network capabilities are allowed as the exam is an individual assessment and no collaboration is allowed. --- ONLINE FINAL EXAM IN CASE OF COVID-19 measures --- In the case of COVID-19 restrictions to on-campus exam, the exam will be online. The exam is an open book exam, the student is allowed to bring all books, lecture notes and worked examples. The students will use their laptops to solve the exam.
Collegerama	No

AESB3341-18	Petrophysics & Image Analysis	5
Responsible Instructor	Dr. K.H.A.A. Wolf	
Instructor	W. Verwaal	
Instructor	J.G. van Meel	
Responsible for assignments	Dr. A. Barnhoorn	
Co-responsible for assignments	Ing. M. Slob	
Contact Hours / Week x/x/x/x	39 lecture and exercise hours, 16 practical hours, 89 self-study hours	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BSc-lectures and practicals in geology, mineralogy, petrology, flow in porous media, mechanics, acoustics, etc. See prerequisites.	
Course Contents	Basics of log evaluation and introduction to measurements on rock. Porosity, permeability, capillarity, parameter determination of rock, principles of acoustic, electric, electro-magnetic and nuclear measurements in boreholes, identification of water, hydrocarbons and coal, quantification of reservoir properties. Lectures and exercises: - Introduction to the identification, measurement and use of physical rock parameters. Rock mineral content and rock textures, of mainly sedimentary rocks, are related to properties like density, conductivity, magnetics, (natural/induced) radiation, acoustics and rock mechanics. - Finally, Physical, chemical and mechanical concepts to define rock content, rock behaviour and the related accuracy of measuring methods, have to be understood. At the end of the lectures and practicals, often used digital log data and interpretation tools are provided to give you the opportunity to create your own log evaluation suite for lithology and content interpretation. These scripts will be used during the Bachelors Field Development Coure(AESB3440. - In relation to the reconstruction of a synthetic mineral content, methods to derive porosity, permeability and capillarity are discussed and calculated in practicals. The practical work consists of four ways to measure permeability and porosity and determination of rock mechanical properties by using a combined acoustic/stress experiment: 1. Laboratory measurement: Darcy-flow and porosity. A. Experiments at the Geoscience and Engineering Laboratory. This work includes measurements on solid and porous cores. Measured are: Bulk volume and bulk weight and the related matrix volume and matrix density; Calculation of the porosity on the two porous samples, with a dry method and the wet/dry method; Calculation of the gas-permeability of the two porous samples; Estimation of the measurement errors and calculation of the accuracy of the measurements. B. Combined acoustic and rock mechanical measurements. Uni-axial stress is put on a solid core. Different types of acoustic waves are used to measure travel time through the undamaged and fractured rock specimen. 2. Empirical relations. A. Carman-Kozeny permeability. Based on spatial distributions of a porous rock texture, porosities and permeabilities are calculated by using statistical spatial characteristics, i.e. grain size distributions, derived volumes and specific surfaces. B. Quantification of grain sizes and grain-size distributions, by using rock surfaces and images. In this exercise the students get familiar with the concept of empirical relations. How are grain-size distribution, porosity, permeability and capillarity connected to each other?	
Study Goals	The student is trained on the essentials regarding interpretation of rock type/mineralogy, related rock physical properties and rock textures. This to define its mineral/fluid composition and geological/mechanical properties. The student learns how to measure and interpret rock characteristics. Finally they learn to do a standard log evaluation on clastic sediments. The obtained experience is essential and compulsory for the Field Development Project (AESB3400) in Period 4.	
Study Goals continuation	At the end of this course, the student is able to: - interpret rock types and mineralogy - understand the underlying rock physics and is able to recognize rock textures. - use multiple methods of measuring rock properties - measure and interpret rock characteristics using laboratory instruments and dedicated software. - perform a simple log evaluation of clastic- and bio-sediments using standard logs. - complete a digital log evaluation and several types of laboratory tests, both alone and in groups. The new knowledge is used in the BSc. Field Development Course in period 4.	
Education Method	Alternating lectures and exercises for an exam plus three series of (laboratory) experiments with reports. - In total 10 to 11 lectures and exercises of 3 hours (1 ECT). - One morning/afternoon an appointment for porosity/permeability measurements with W. Verwaal (1 ECT), and, - One morning/afternoon an appointment for acoustic/rock mechanics measurements with A. Barnhoorn and W. Verwaal, and (1 ECT), - Two mornings/afternoons for texture analysis and Carman-Kozeny permeability with K-H. Wolf and J. van Meel (2 ECT). Appointments for the practical work are to be made with: Wim Verwaal: tel. 015 278 1326 Joost van Meel: tel. 015 278 4986 All reports are to be sent to: ' imageanalysis-CITG@tudelft.nl ' or a new address (to be announced), before a given deadline.	
Computer Use	Yes; for the exercises and the final log-evaluations, a laptop and/or the TU-Delft desktops are needed.	
Course Relations	- All BSc-related geology topics. - Mechanics and Transport by Flows in Porous Media (AESB3340), - Field Exploration Project (AESB3440)	
Literature and Study Materials	Lecture notes (IMPORTANT and under constant revision), slides, practicals with answerd, related literature and excerpts from handbooks.	
Prerequisites	mineralogy and geology (AESB1130, AESB1230), Sedimentology & Reservoir Geology(AESB2230), Physical Transport Phenomena (AESB2320)	
Assessment	- AESB3341-18, TOETS 1: Final exam about lectures and exercices (1 EC). - AESB3341-18, TOETS 2: Two reports from laboratory experiments, to measure porosity, permeability, acoustics and rock mechanics properties (2 EC). - AESB3341-18, TOETS 3: Two reports on porosity and permeability based on image analysis data and empirical/spatial/hybrid relations (2 EC). So, groups of 2 students prepare four reports and send those before a given deadline to: 'imageanalysis-CITG@tudelft.nl'. or a new address (to be announced).	

	Judgement: - Written exam:1 ECTS - Porosity/Permeability practical by using image analysis: 1 ECTS - Porosity/Permeability practical by using spatial characteristics: 1 ECTS - Porosity/Permeability practical in the laboratory: 1 ECTS - Rock mechanics/acoustical practical in the laboratory: 1 ECTS
Permitted Materials during Tests	Required: Ruler, simple pocket calculator, color pencils. Not allowed: cell phone, cameras, sophisticated calculators.
Tags	Analysis Design Energy Geo Engineering Group work Image processing Lab Research Personal Physics Practicals Signals Small groups Software Underground
Contact	Dr. K-H. A.A. Wolf k.h.a.a.wolf@tudelft.nl tel: +31 (0)15 27 86029 Dr. A.Barnhoorn Auke.Barnhoorn@tudelft.nl tel: +31 (0)152789682 Teaching instructors: Joost van Meel j.g.vanmeel@tudelft.nl tel: +31 (0)6 48875959 Wim Verwaal W.Verwaal@tudelft.nl tel: +31 (0)152781326 Guus Lohlefink C.G.K.Lohlefink@tudelft.nl tel: +31 (0)2787663
Expected prior Knowledge	BSc-lectures and practicals in geology, mineralogy, petrology, flow in porous media, mechanics, acoustics, etc. See prerequisites.
Academic Skills	The student is able: - to follow lectures and understand the lecture notes. - to perform practicals related to the lectures and make basic log evaluations. - to perform experimental work in a laboratory and write practical reports.
Literature & Study Materials	- Petrophysics Lecture notes and overhead sheets by K-H. Wolf. - Excerpts from log-evaluation manuals. - Excercises related to the lecture notes. - Introductions to the four practicals and associated relevant literature.
Judgement	- Written exam:1 ECTS - Porosity/Permeability practical by using image analysis: 1 ECTS - Porosity/Permeability practical by using spatial characteristics: 1 ECTS - Porosity/Permeability practical in the laboratory: 1 ECTS - Rock mechanics/acoustical practical in the laboratory: 1 ECTS For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Accepted during the written exam: Ballpoints, colour pencils, ruler, triangle, pocket calculato, eraser, and other relevant office supply. Strictly forbidden: Mobiles, computers or other equipment that can make and send or receive images and text.
Collegerama	No

AESB3341-18 Toets 1	Exam	1
Responsible Instructor	Dr. K.H.A.A. Wolf	
Instructor	W. Verwaal	
Instructor	J.G. van Meel	
Co-responsible for assignments	Dr. A. Barnhoorn	
Co-responsible for assignments	Ing. M. Slob	
Contact Hours / Week x/x/x/x	see description AESB3341-18	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	see description AESB3341-18	
Study Goals	see description AESB3341-18	
Education Method	see description AESB3341-18	
Assessment	see description AESB3341-18	
Expected prior Knowledge	see description AESB3341-18	
Academic Skills	see description AESB3341-18	
Literature & Study Materials	see description AESB3341-18	
Judgement	see description AESB3341-18	
Permitted Materials during Exam	see description AESB3341-18	
Collegerama	No	

AESB3341-18 Toets 2	Lab report	2
Responsible Instructor	Dr. K.H.A.A. Wolf	
Instructor	W. Verwaal	
Instructor	J.G. van Meel	
Responsible for assignments	Dr. A. Barnhoorn	
Co-responsible for assignments	Ing. M. Slob	
Contact Hours / Week x/x/x/x	see description AESB3341-18	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	see description AESB3341-18	
Study Goals	see description AESB3341-18	
Education Method	see description AESB3341-18	
Assessment	see description AESB3341-18	
Expected prior Knowledge	see description AESB3341-18	
Academic Skills	see description AESB3341-18	
Literature & Study Materials	see description AESB3341-18	
Judgement	see description AESB3341-18	
Permitted Materials during Exam	see description AESB3341-18	
Collegerama	No	

AESB3341-18 Toets 3	*Image Analysis*	*2*
Responsible Instructor	Dr. K.H.A.A. Wolf	
Instructor	W. Verwaal	
Instructor	J.G. van Meel	
Co-responsible for assignments	Ing. M. Slob	
Co-responsible for assignments	Dr. A. Barnhoorn	
Contact Hours / Week x/x/x/x	see description AESB3341-18	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	see description AESB3341-18	
Study Goals	see description AESB3341-18	
Education Method	see description AESB3341-18	
Assessment	see description AESB3341-18	
Expected prior Knowledge	see description AESB3341-18	
Academic Skills	see description AESB3341-18	
Literature & Study Materials	see description AESB3341-18	
Judgement	see description AESB3341-18	
Permitted Materials during Exam	see description AESB3341-18	
Collegerama	No	

AESB3343	Mineral Resource Geology and Modelling	5
Responsible Instructor	Dr. M.W.N. Buxton	
Contact Hours / Week x/x/x/x	Total 28 x 5=140 hours. Oral lectures: 56 contact hours	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	- Mineral resources: fundamental need for resources for society, growing demand for resources for the energy transition, maximal use of resources/resource efficiency, global supply and demand and sustainability and circularity of supply. - What: Ore deposit styles o Reference will be made to the following specific commodities: X Copper, Iron X Energy transition: Overview on required commodities, specifically Platinum group elements, Critical Raw Materials (Rare Earth Elements, Cobalt, Lithium). - How: Exploration and discovery methods - Where: the mining value chain: o Exploration, o Resource Definition/modelling, o Mining, o Mineral processing, o Final product, o Circularity and recycling. - Why: Importance of Resource Modelling and Estimation in the Value Chain of Mining, - The use of computer-aided tools for resource modelling o Explorative Data Analysis, o Model Assumptions of Stationarity and Ergodicity, o Variography and Variogram Modelling of mineral resources, o Geostatistical Methods for Spatial Estimation including Simple Kriging and Ordinary Kriging for application of mineral resource modelling. - Grade Tonnage Curves, with reference to real practical examples, - Introduction to CRIRSCO-based International Reporting standards (example JORC Code).	
Study Goals	By the end of this course, students will be able to: - recognise characteristics, styles and features of occurrences of specific commodity types (Copper, Iron, Platinum group elements, Critical Raw Materials (Rare Earth Elements, Cobalt, Lithium)) of relevance to society; - explain the importance of mineral raw materials to enable the energy transition; - evaluate the economic potential of the specific commodities for society; - summarise the societal requirements for continued supply of raw materials from primary and secondary sources; - define exploration methods for specific commodities (Copper, Iron, Platinum group elements, Critical Raw Materials (Rare Earth Elements, Cobalt, Lithium)); - explain the importance for Mineral Resource modelling and estimation and its contribution to the mining value chain; - explain basic terminology related to Mineral Resources and Reserves (JORC Code (see the course description) internationally accepted terminology for description of mineral resources); - identify suitable methods for spatial modelling of ore body attributes related to the problem on hand by applying methods of explorative data analysis and variography; - judge assumptions in resource modelling, confidence in resultant models and their inherent uncertainty; - assess the significances/consequences for real-world examples of mineral resource models; - perform a computer-aided resource estimation exercise for a geologically continuous orebody from drill hole data to reporting on resources using commercially available software.	
Education Method	Combination of lectures, self-study and group work --- UPDATE dd 2-2-2021 due to the COVID-19 measures --- Combination of online lectures, self-study and group work	
Assessment	- Study load: Total 28 x 5=140 hours. Oral lectures: 56 contact hours, - Teaching method(s): Lectures, assignments, practical work in projects of 2 to 4 students, self-studying, - Literature & study material: The slides will be provided digitally, references for reading are provided. - Assessment method(s): The assessment consists of a set of individual assignments (20%) a written exam (40%) and a report of a team project (40%). - Means permitted during exam: only printed materials and table calculator, worked out questions are not allowed. - Grading: Written exam 2/5, report 2/5 and assignments 1/5. --- UPDATE dd 2-2-2021 due to the COVID-19 measures --- The assessment consists of a set of individual assignments (20%) a written exam (with an online exam as remote alternative) (40%) and a report of a team project (40%). In order to pass the course, the minimum grade for each part should be at least 6.0	
Expected prior Knowledge	Statistics, Geostatistics & Remote Sensing course, Geology basics	
Academic Skills	Explain, assess, judge.	
Literature & Study Materials	Powerpoint slides, recommended journal papers and texts	
Judgement	Written exam 2/5, report 2/5 and assignments 1/5 For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Only printed materials and table calculator, worked out questions are not allowed.	

AESB3343 Toets 1	Assignments	1
Responsible Instructor	Dr. M.W.N. Buxton	
Contact Hours / Week x/x/x/x	See AESB3343	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	See AESB3343	
Study Goals	See AESB3343	
Education Method	See AESB3343	
Assessment	See AESB3343	
Expected prior Knowledge	See AESB3343	
Academic Skills	See AESB3343	
Literature & Study Materials	See AESB3343	
Judgement	See AESB3343	
Permitted Materials during Exam	See AESB3343	

AESB3343 Toets 2	Exam	2
Responsible Instructor	Dr. M.W.N. Buxton	
Contact Hours / Week x/x/x/x	See AESB3343	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	See AESB3343	
Study Goals	See AESB3343	
Education Method	See AESB3343	
Assessment	See AESB3343	
Expected prior Knowledge	See AESB3343	
Academic Skills	See AESB3343	
Literature & Study Materials	See AESB3343	
Judgement	See AESB3343	
Permitted Materials during Exam	See AESB3343	

AESB3343 Toets 3	Project Report	2
Responsible Instructor	Dr. M.W.N. Buxton	
Contact Hours / Week x/x/x/x	See AESB3343	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	See AESB3343	
Study Goals	See AESB3343	
Education Method	See AESB3343	
Assessment	See AESB3343	
Expected prior Knowledge	See AESB3343	
Academic Skills	See AESB3343	
Literature & Study Materials	See AESB3343	
Judgement	See AESB3343	
Permitted Materials during Exam	See AESB3343	

AESB3400	Bachelor Thesis	10
Responsible Instructor	Dr. P.B.R. Bruna	
Course Coordinator	Y. de las Heras	
Course Coordinator	Dr. P.B.R. Bruna	
Contact Hours / Week x/x/x/x	Scheduled on individual basis	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	The course content depends on the actual project and is described separately for each project on a yearly basis in Brightspace	
Study Goals	A student has to choose an individual Geoscience - curriculum related - research project The project is submitted as an assignment (definition of your project form) in Brightspace. The form is evaluated by Dr. Bruna during the first week of the BEP in order to verify that it complies with the basic study goal and with supervision requirements. During the Bachelor End Project [BEP], the graduate student learns to apply her/his skills and knowledge gained in their bachelor study, which she/he has to carry out independently. The grade given after the defense by the BEP assessment committee (Art. 23, board of examiners [BoE] https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/onderwijsinformatie/educational-rules-and-regulations), will be validated after handing in a self-evaluation, the power point presentation and the plagiarism check (obtained in Ouriginal by the supervisor) by the student. After this, and after the assessment rubric is uploaded in OnStage, the administrative coordinator will enter the grade in Osiris.	
Study Goals continuation	After the Bachelor Thesis a student is able to: 1. formulate a research question and hypothesis on the requested abstraction level based on a project description and develop a substantiated and systematic approach for the research project. 2. contribute to scientific and/or engineering knowledge by conducting research. They gather, synthesize, and interpret data based on research questions. They use known and/or new research techniques and the knowledge obtained in the AES bachelor curriculum. 3. formulate validated conclusions based on discussion of the results by using valid and defendable arguments. They provide relevant recommendations for the project. They include a discussion regarding the environmental, economic, social and ethical consequences. 4. make a feasible project planning no later than week 2 of the BEP (Gantt-chart) and motivates the time spent on the activities. They execute the project as an individual and show initiative and independency. 5. produce a structured and consistent report following an academic standard. 6. orally present the research project within thirty minutes in a concise manner to supervisors, reviewers and public. The student can discuss and answer questions correctly. 7. reflect on their working process and reflect on their learning process by elaborating on the mastery of the learning objectives.	
Education Method	The course consists of a project with guidance from a team of two official supervisors. The BEP have a limited supervision; each student is expected to work mostly on her/his own. Supervision (opposed to assessment committee): the guidance of the BEP is provided by two scientists (possibly from the same section) appointed as a supervisor by the BoE (the list of the official supervisors is available in Onstage). Post-doc or PhD, who have not been appointed as supervisor by the BoE can also be added unofficially in the supervision team.	
Course Relations	All courses in the bachelor program are related to this course.	
Literature and Study Materials	Many project descriptions are available on Brightspace. Other projects can be decided between a student and her/his potential supervisors. If the student have difficulties to find a project, she/he can contact Dr. Bruna	
Prerequisites	First year and at least 60 EC from the second and third year curriculum of the BSc program Applied Earth Sciences, excluding the minor. For all requirements, forms and agreements, please read the manuals on Brightspace and if necessary contact Mrs. Yolanda de las Heras - studentoffice-esa-ceg@tudelft.nl. Students should register themselves in OnStage. (its the same as like registering for an exam) The data and deadline for registration are published on Brightspace AESB3400 Q1 (Q2, Q3, Q4) and on the general page for Bachelor students AES / CEG.	
Assessment	The student writes a bachelor thesis report assessed by at least one of her/his official supervisors and by an external assessor (not a supervisor) chosen to sit in the assessment committee (Art 23). The draft version(s) (one week before the defense date) and the final version of the BEP thesis must be uploaded as official assignments into OnStage. The final assignment will be checked for plagiarism (Ouriginal) by the student or by the supervisor before the defense (requirement to be checked in Brightspace Bachelors Graduation Project Manual). The student has to send the report to the assessment committee for the plagiarism to be checked before the defense (thresholds are available into Brightspace). At the end of the project and within the teaching period, the student gives an oral presentation for a general academically educated audience (the defense is public). It is of student responsibility to contact and constitute his assessment committee in close discussion with her/his supervisors. After the defense, the student answers questions from the audience. The student is then invited to answer specific questions about her/his research (for 30-60 minutes) in front of the assessment	

	committee (without the presence of the audience). The assessment committee will deliberate without the candidate and evaluate the work following a grading grid which is common to all BEP theses. The categories for which the student is assessed are briefly explained below. After the deliberation, the final grade is communicated publically to the student. After the defense the student copies the link of the plagiarism check into a blank page and upload it into OnStage
Contact	Dr. Pierre-Olivier Bruna p.b.r.bruna@tudelft.nl Mrs. Yolanda de las Heras Y.delasHeras@tudelft.nl
Expected prior Knowledge	First year and at least 60 EC (excluding the minor) from second year and third year of the BSc program Applied Earth Sciences. The topics needed for your assignment must have been attended and at least one time been examined.
Academic Skills	See: Study Goals continuation
Literature & Study Materials	Depending on the topic.
Judgement	Judgement (opposed to supervision): following Art. 23, the student work is judged by at least one of the two official supervisors and a second reviewer from another section who is entitled to judge educational assignments (the list of requirements is provided in Art. 23). The students research is judged following six rubrics: research approach (10%), profoundness (30%), evaluation of the results (20%), organization (5%), written report (25%) and oral presentation (10%). The description of the various categories is available upon request. The rubrics with the motivation of the judgement must be signed by the reviewers before submitting the result to the Student Office ESA CEG - Studentoffice-esa-ceg@tudelft.nl. The rubrics are uploaded into OnStage by the reviewers. For more information on grading, see article 14 in the Educational Rules and Regulations: https://www.tudelft.nl/studenten/faculteiten/citg-studentenportal/onderwijs/onderwijsinformatie/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

AESB3440	Field Exploration Project	5
Responsible Instructor	Dr. A.M.H. Pluymakers	
Module Manager	Dr. A.M.H. Pluymakers	
Module Manager	Ir. M. Keersemaker	
Instructor	Ir. F.C. Vossepoel	
Instructor	Dr. K.H.A.A. Wolf	
Instructor	Dr. R. Ghose	
Instructor	Ir. J.R. van Duijvenbode	
Instructor	Dr. H.A. Abels	
Instructor	Dr.ir. A.A.M. Dieudonné	
Contact Hours / Week x/x/x/x	The total study load is 140 hours.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course content depends on the actual projects and is described during the introduction separately for each project on a yearly basis	
Study Goals	At the end of this course you will be able to design a subsurface project in a designated area with its specific needs and constraints together with a team, and explain why the choices made during the case study represent the best course of action given the physical, environmental, economic and societal constraints with a written report and oral presentation	
Study Goals continuation	By the end of this course, you will also be able to: - Locate data relevant to the technical, environmental, economic, societal, and ethical aspects of a subsurface project - Assess the quality of the gathered data relevant to the technical, environmental, economic, societal, and ethical aspects of a subsurface project - Divide the interdisciplinary complex task related to geo-engineering evaluation in manageable units and to work according to a tight time schedule; - Apply appropriate and innovative engineering skills to solve problems encountered during the project; - Show you can work in a team not of their own choosing and reflect on their own work and that of their partners.	
Education Method	The course consists of a project with short informal lectures given to refresh some material from previous courses and to provide limited new material. The project runs with limited supervision; most of the time the students are working within their assigned groups. This is a deskjob. When on campus, this means 8 hr working days in a classroom with designated lunch breaks. When online, this means 8 hr working days in a digital breakout room with designated lunch breaks and lecture times.	
Course Relations	Engineering design is the process of devising a system, component, or process to meet desired needs. It is a decision-making process (often iterative), in which the basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet these stated needs. Therefore, all courses in the bachelor program are related to this course.	
Literature and Study Materials	To be provided and discussed in the individual project description. Students are also encouraged to search relevant literature and other information by their own in order to improve the quality of work.	
Prerequisites	Fulfilled the first two years of BSc program Applied Earth Sciences; followed courses of Year 3 up to Q3. In exceptional case a student can request the responsible instructor for a deviation from this rule.	
Assessment	The work attitude is assessed during the project; knowledge and skills are assessed during the presentations and evaluation of project report. The project coach and tutor will be accessible during the whole project time and will assess the achievements of learning objectives, the process, skills applied, teamwork as well as the project report and final presentation. A board of experts (partly external) will assess the report and presentation. Using peer review, students will assess the team members and their contribution to the teamwork (formative and summative).	
Contact	Anne Pluymakers anne.pluymakers@tudelft.nl	
Expected prior Knowledge	Courses from the BSc up to year 3, quarter 3.	
Academic Skills	Hard skills: geology, engineering, mathematics, physics, Excel, programming (Matlab or Python) Soft skills: scientific writing, presentation, discussion, team work	
Literature & Study Materials	Will be provided	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	The presentations and presented material are prepared within the assigned groups by making use of available (online) information. Consulting other groups is not allowed.	
Collegerama	No	

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